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(54) **METHOD FOR ADJUSTABLY CLOSING TISSUE**

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Related U.S. Application Data

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Publication Classification

(51) **Int. Cl.**

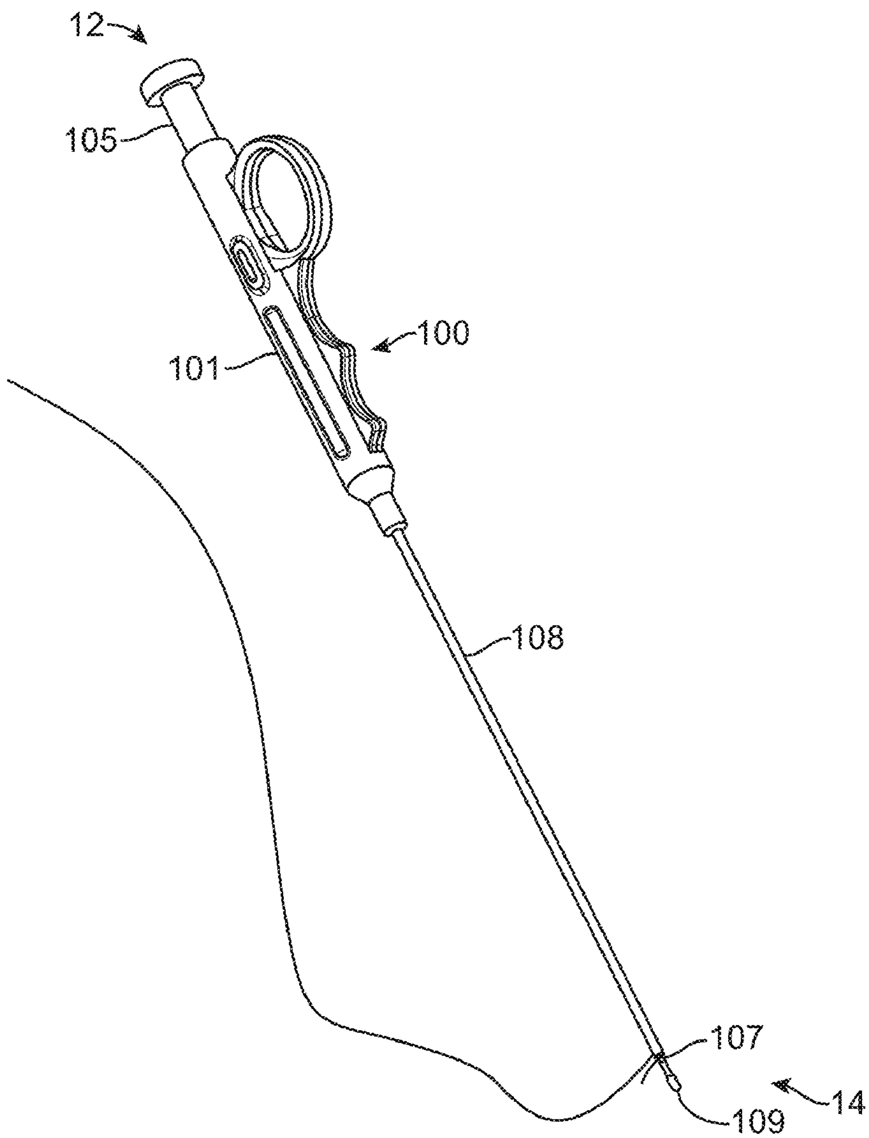
A61B 17/04 (2006.01)

(52) **U.S. Cl.**

CPC *A61B 17/0482* (2013.01); *A61B 17/0469* (2013.01); *A61B 17/0485* (2013.01)

(57) **ABSTRACT**

A system for closing trocar wound sites includes a suture engagement device and a guide to direct the device through the body tissues. An improved needle and guide instrument set allows for the surgical placement of suture to be completed "blind," namely, without the aid of an endoscope for direct visualization of the abdominal cavity. The guide can include markings to accurately determine a depth of the guide and, therefore, accurately determining a suture fascial tissue bite distance. This may be advantageous in certain surgical procedures where an endoscope is not being used or does not provide adequate visualization of the surgical site.



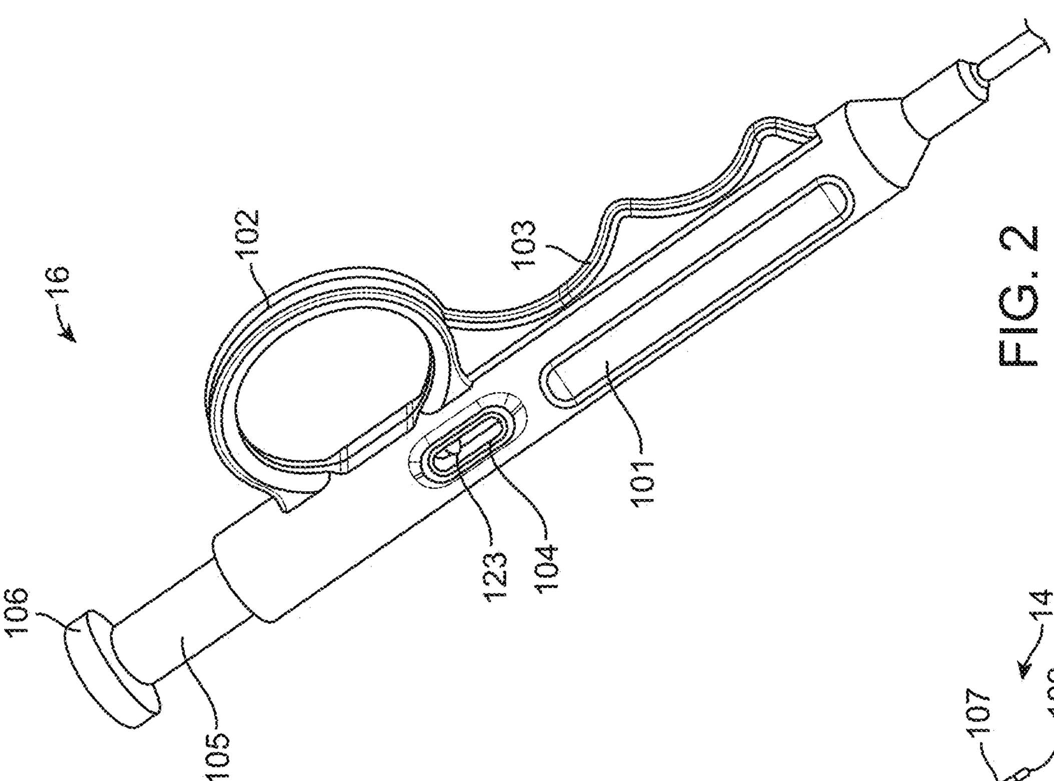


FIG. 2

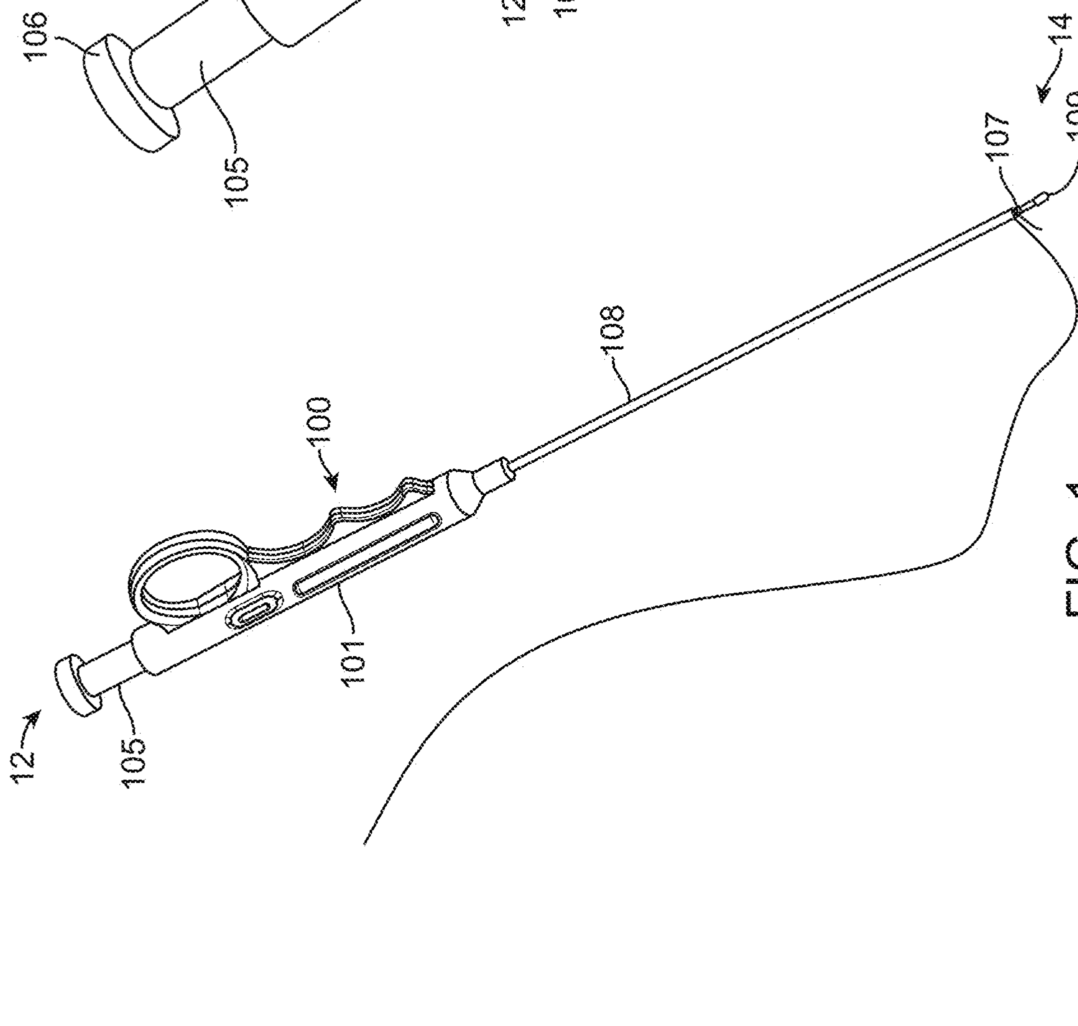


FIG. 1

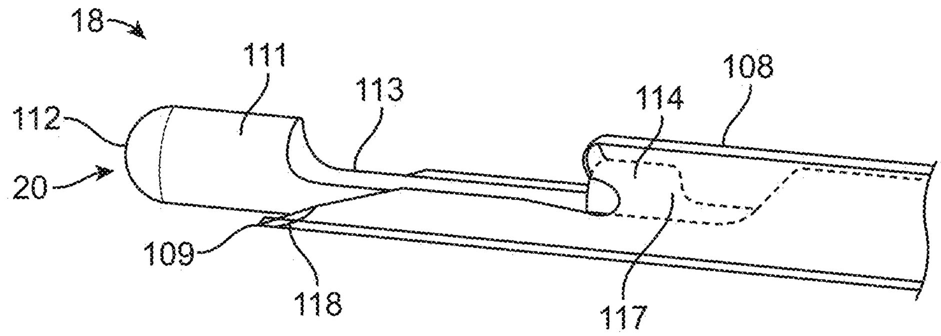


FIG. 3

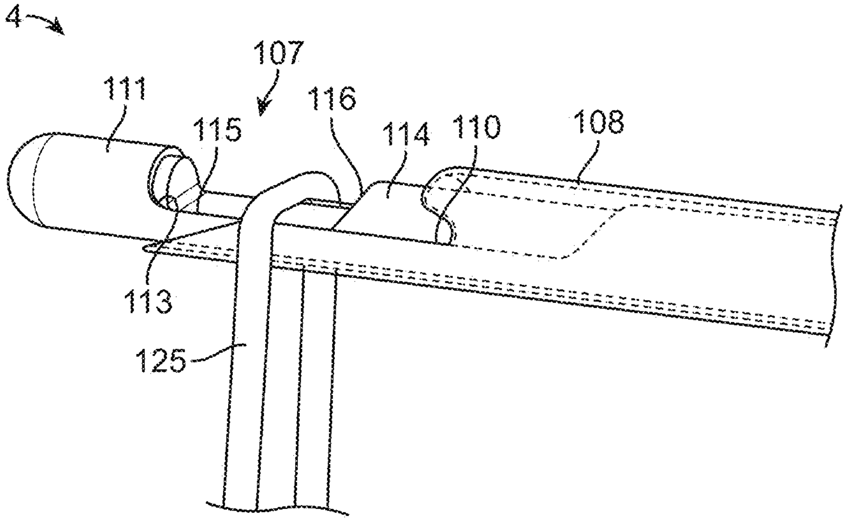


FIG. 4A

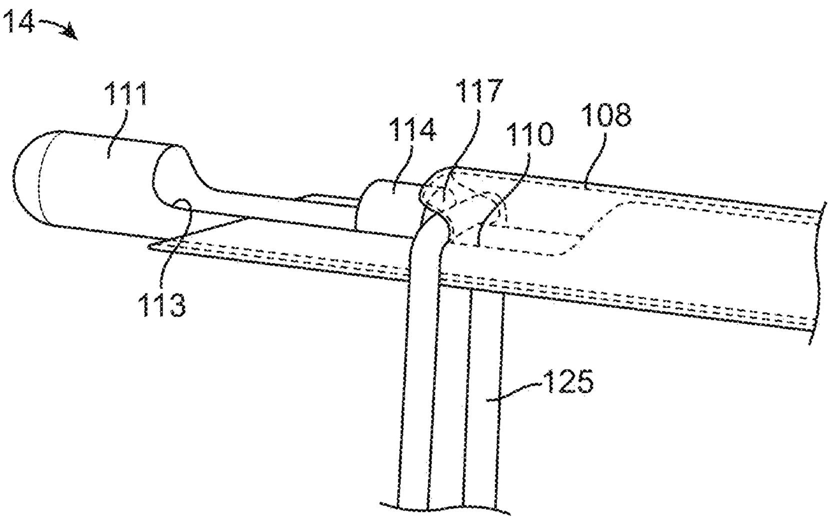


FIG. 4B

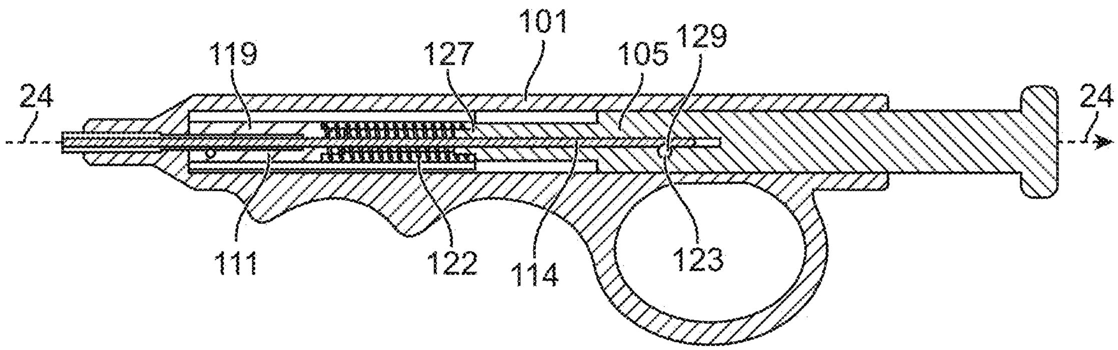

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FIG. 5A

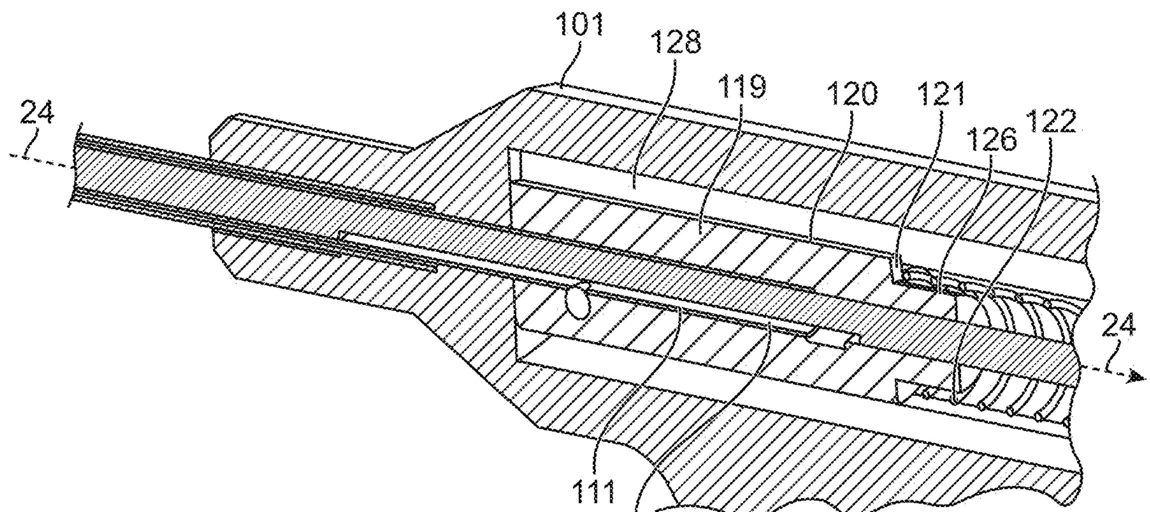
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FIG. 5B

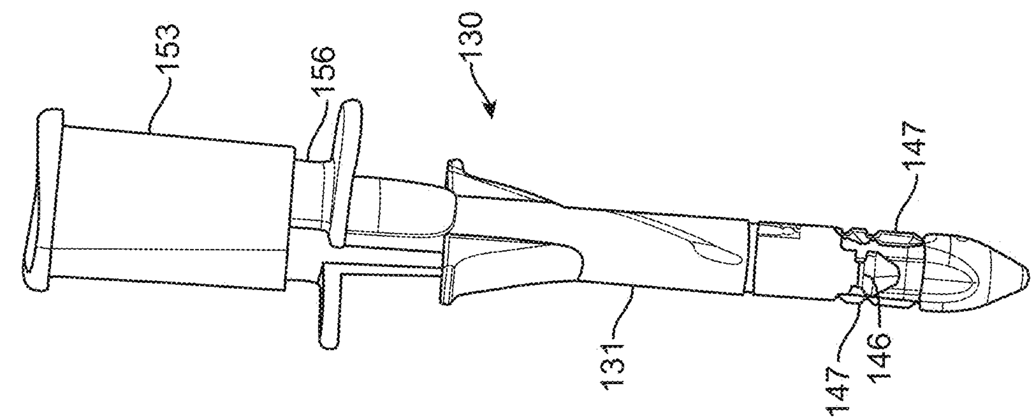

FIG. 6C

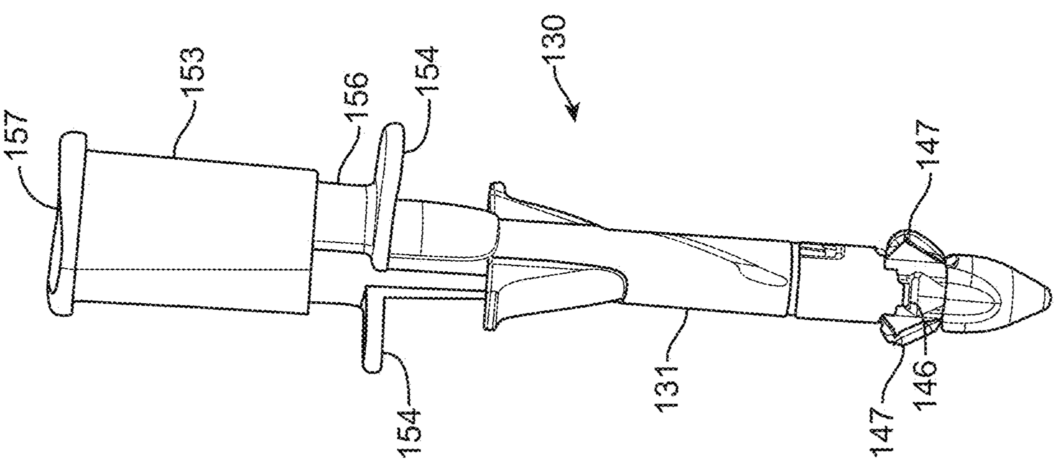

FIG. 6B

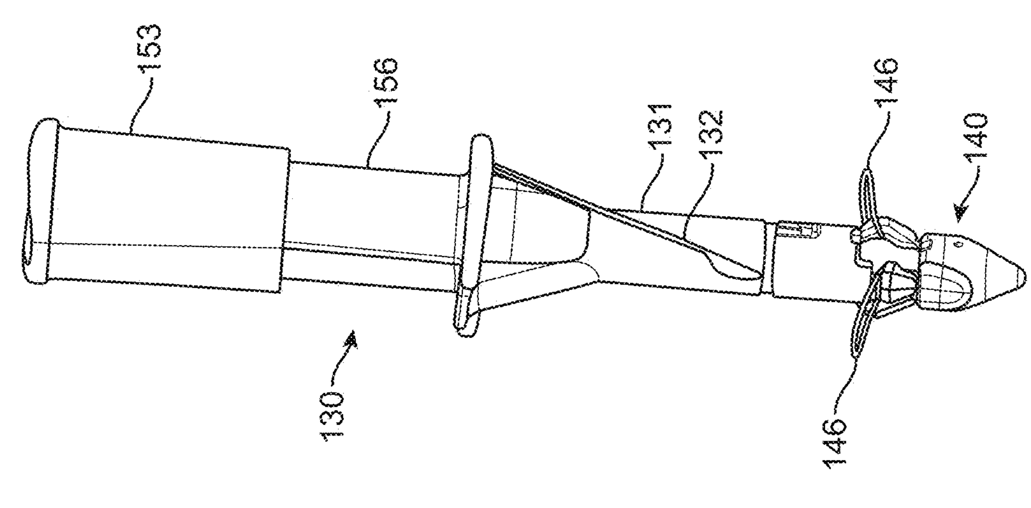

FIG. 6A

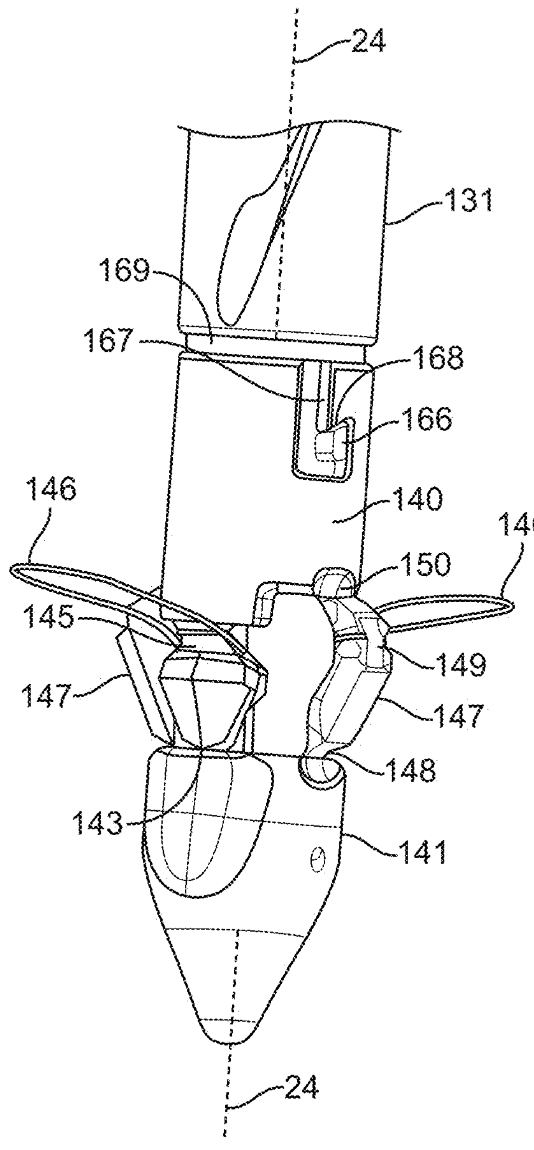
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FIG. 7

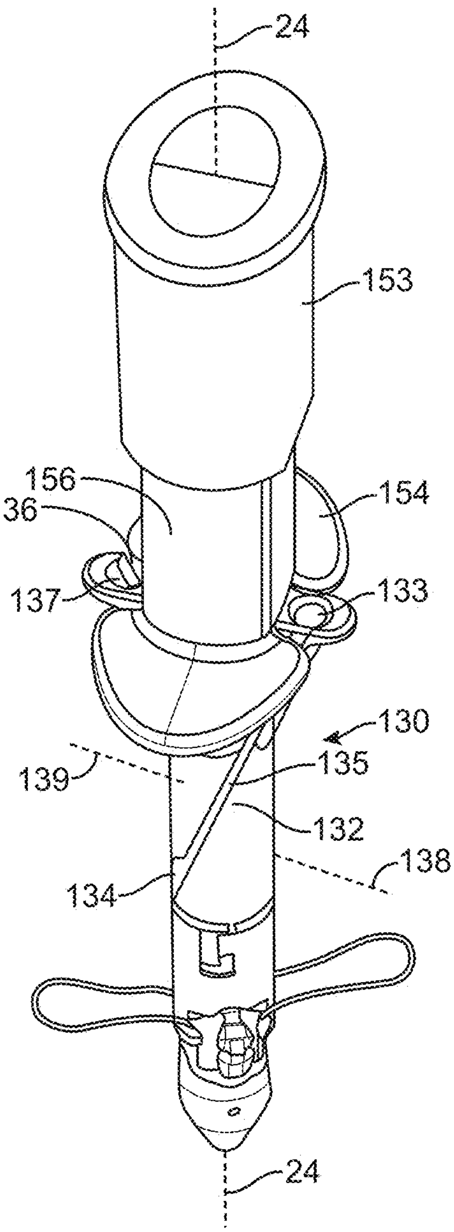
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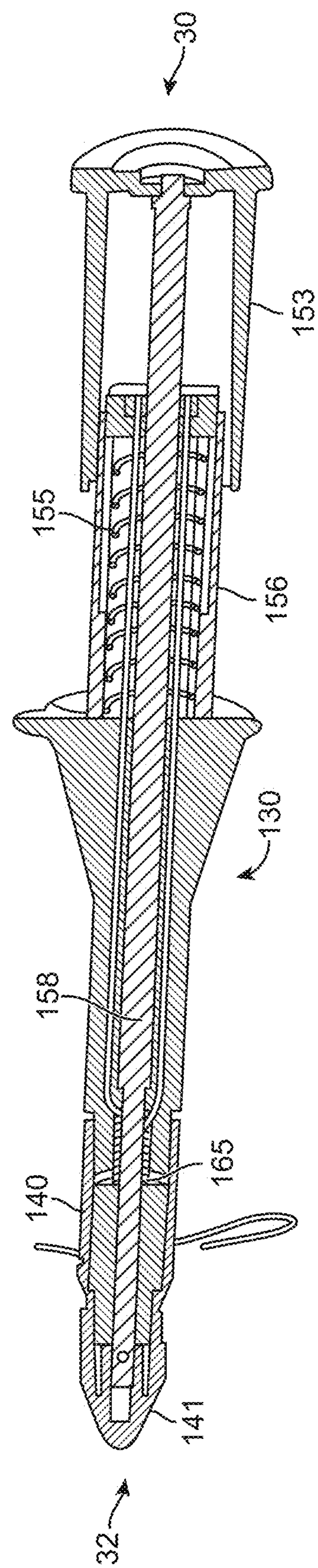
FIG. 8



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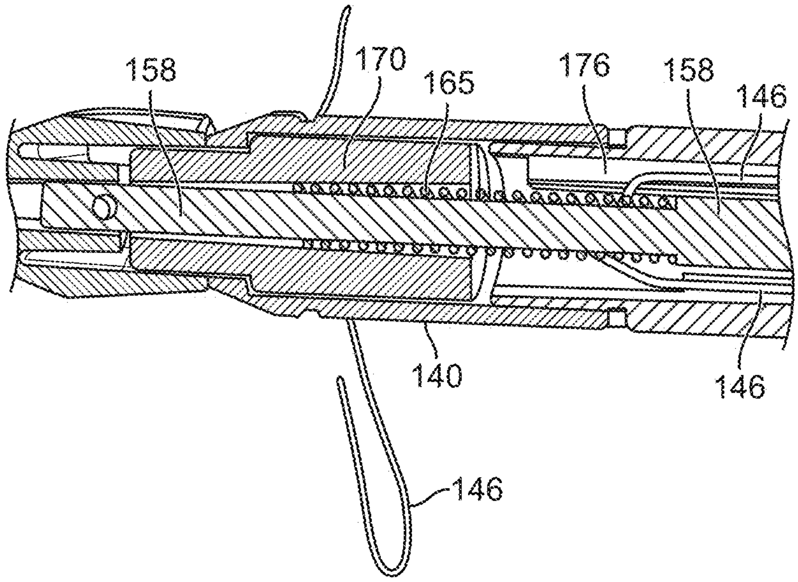
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FIG. 9B

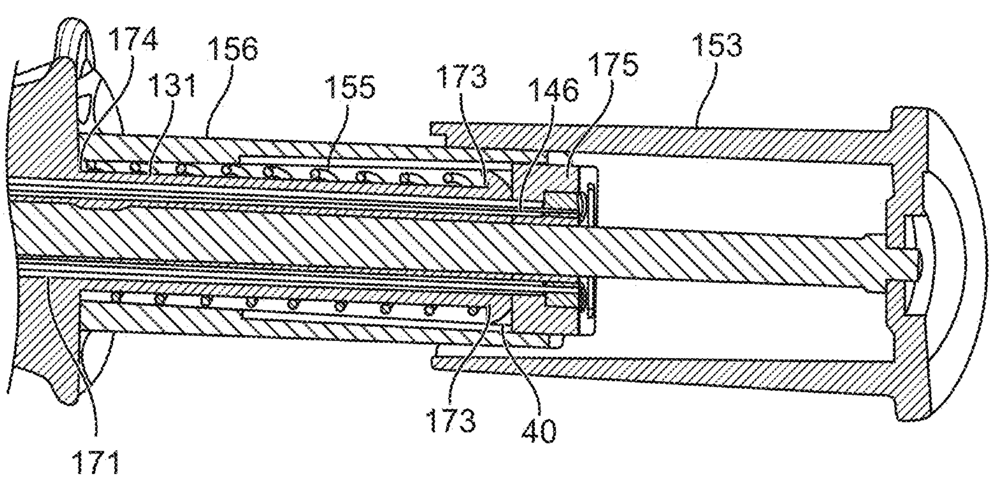
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FIG. 9C

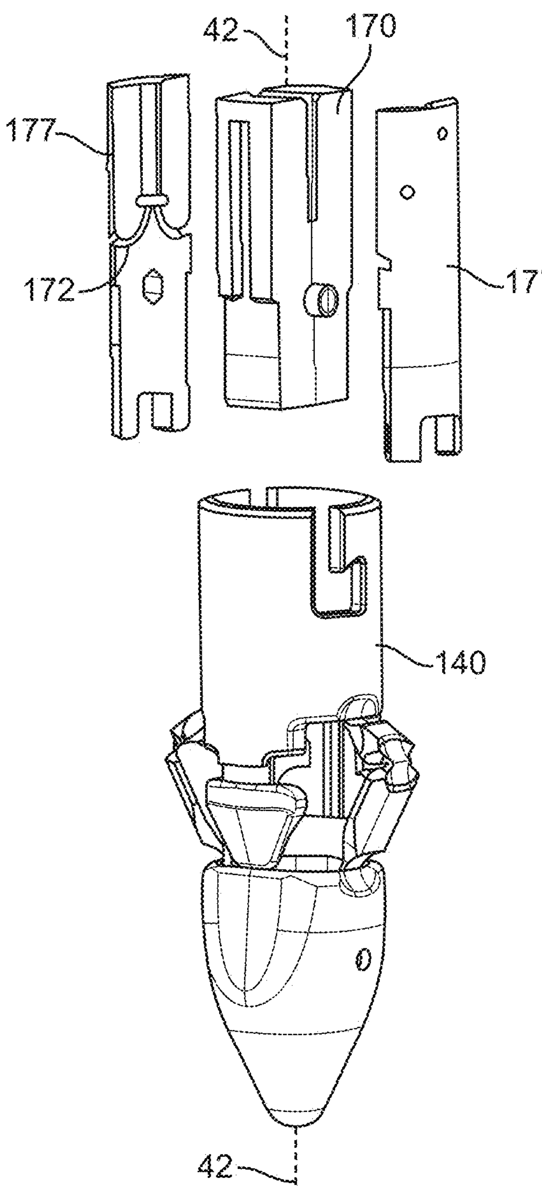
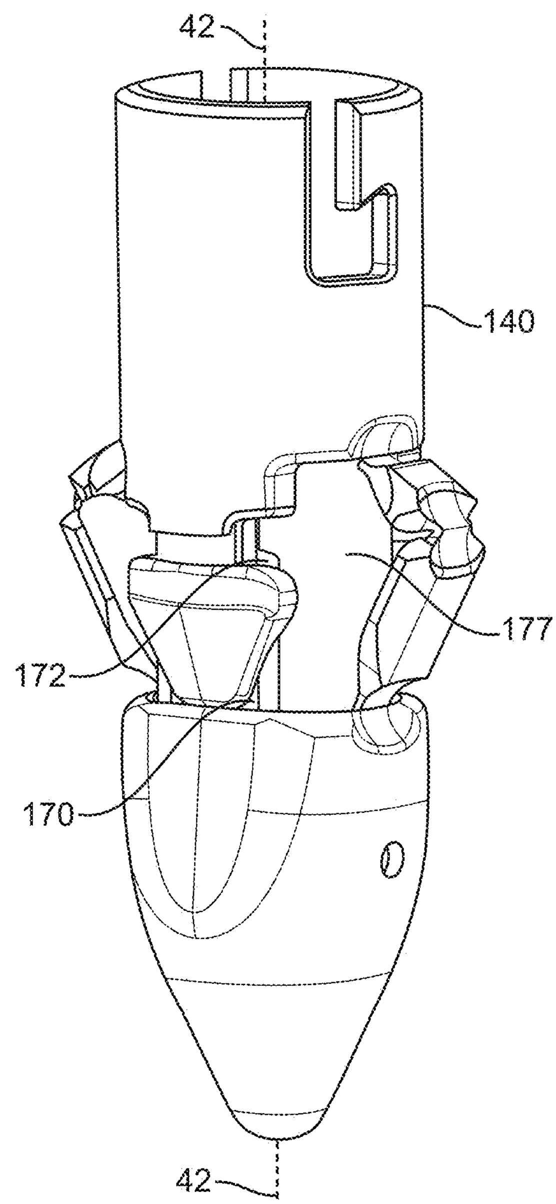


FIG. 10A



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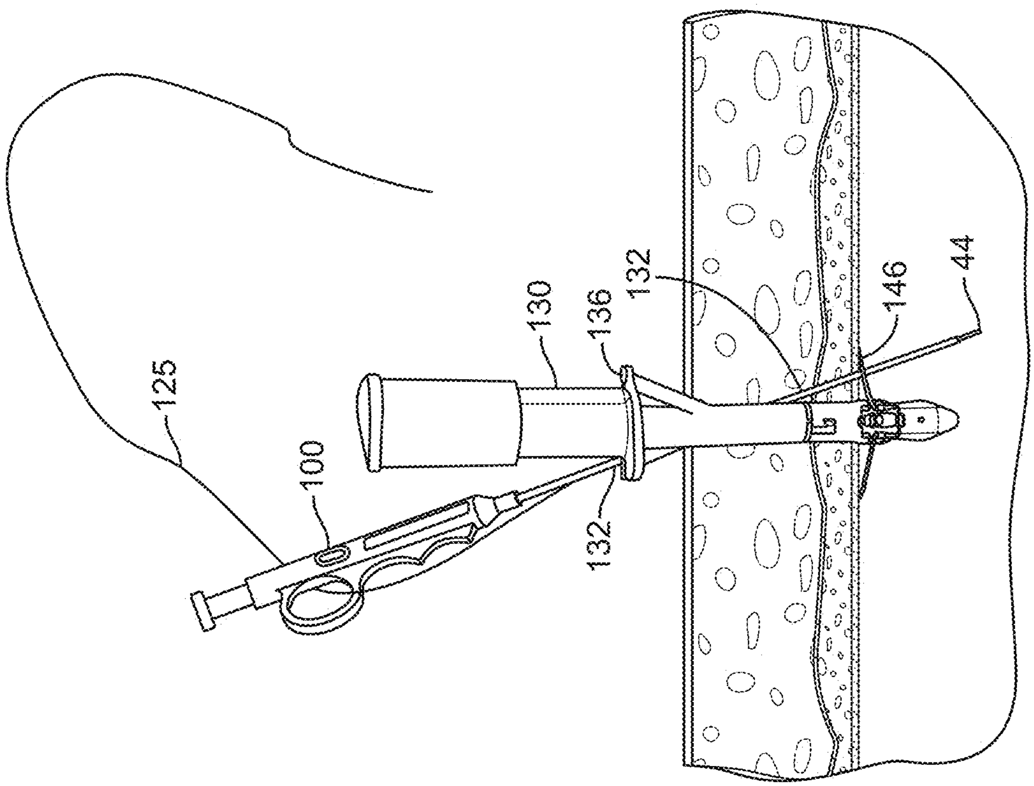


FIG. 12

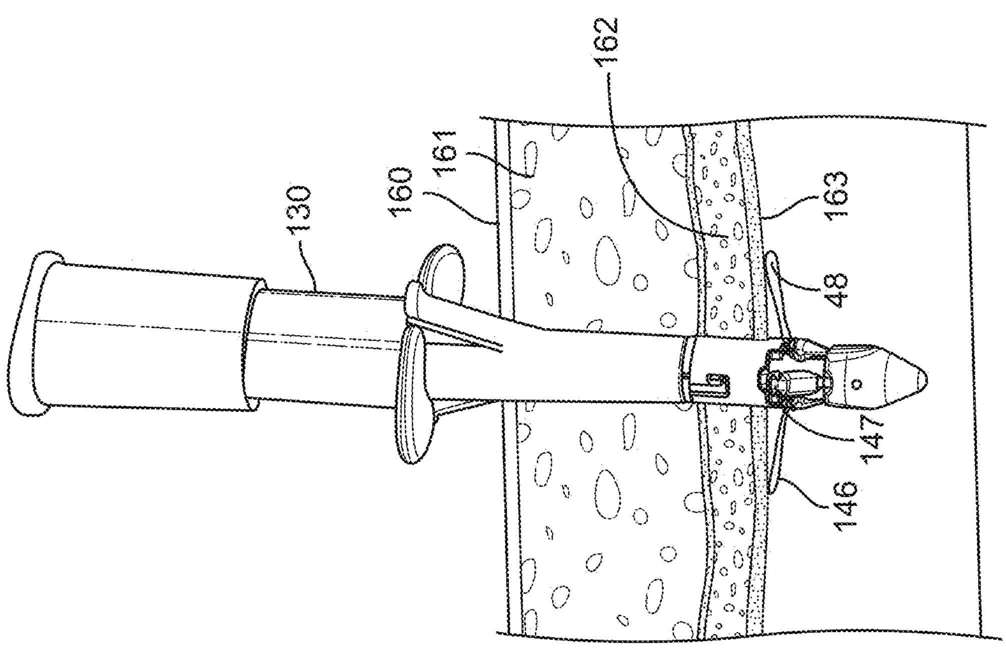


FIG. 11

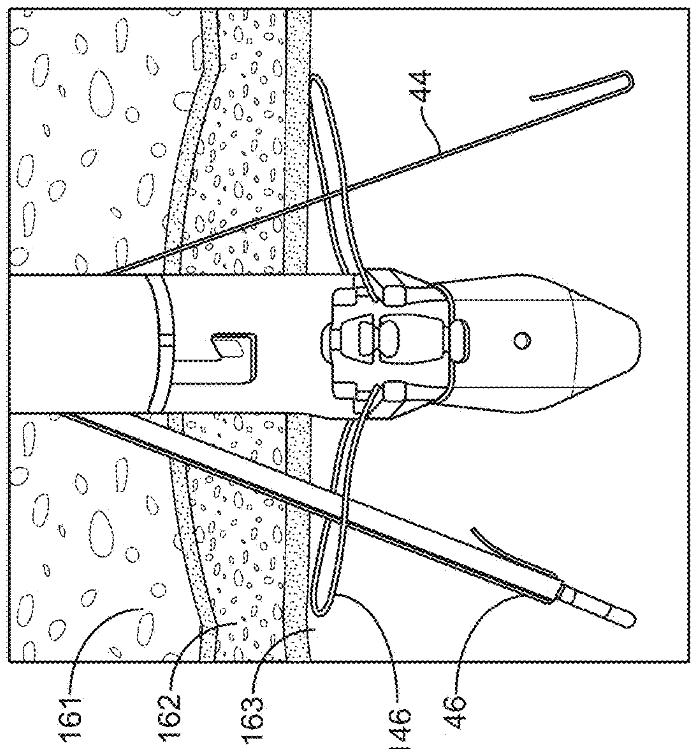


FIG. 13B

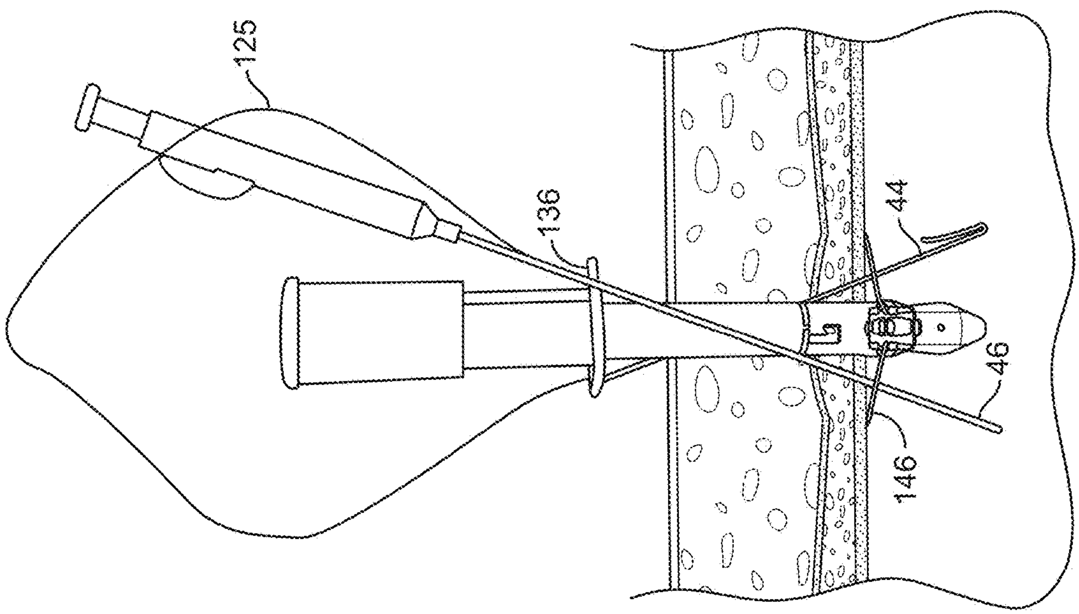
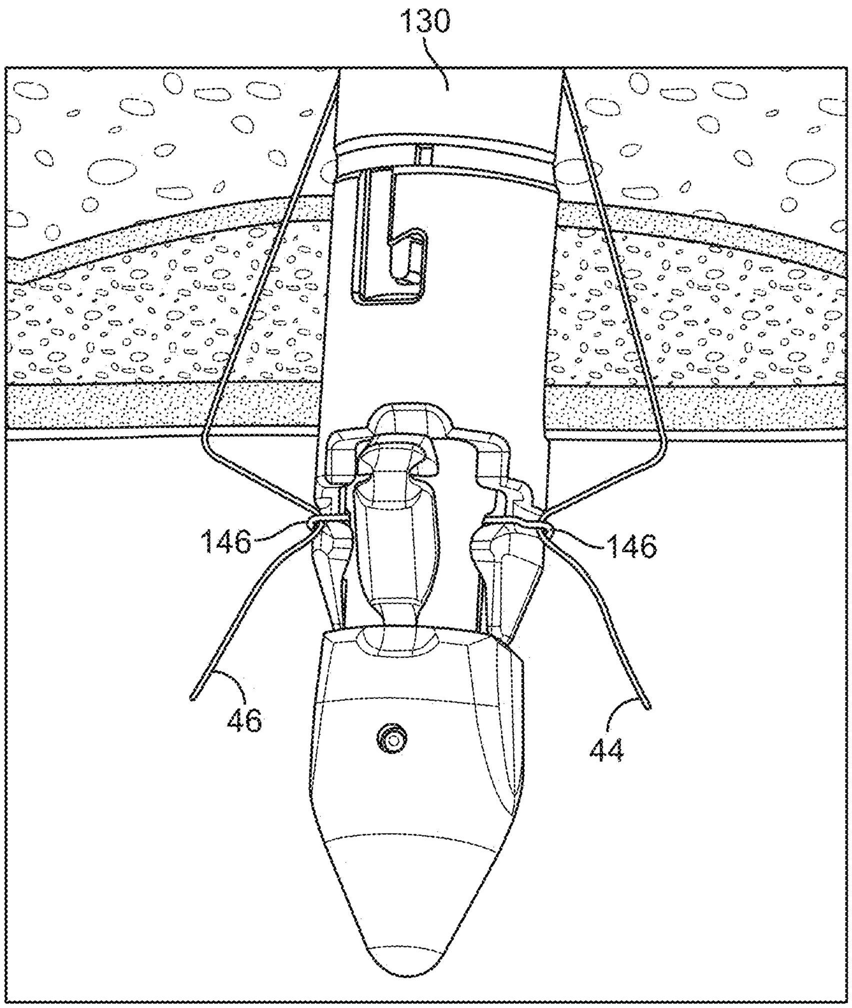


FIG. 13A



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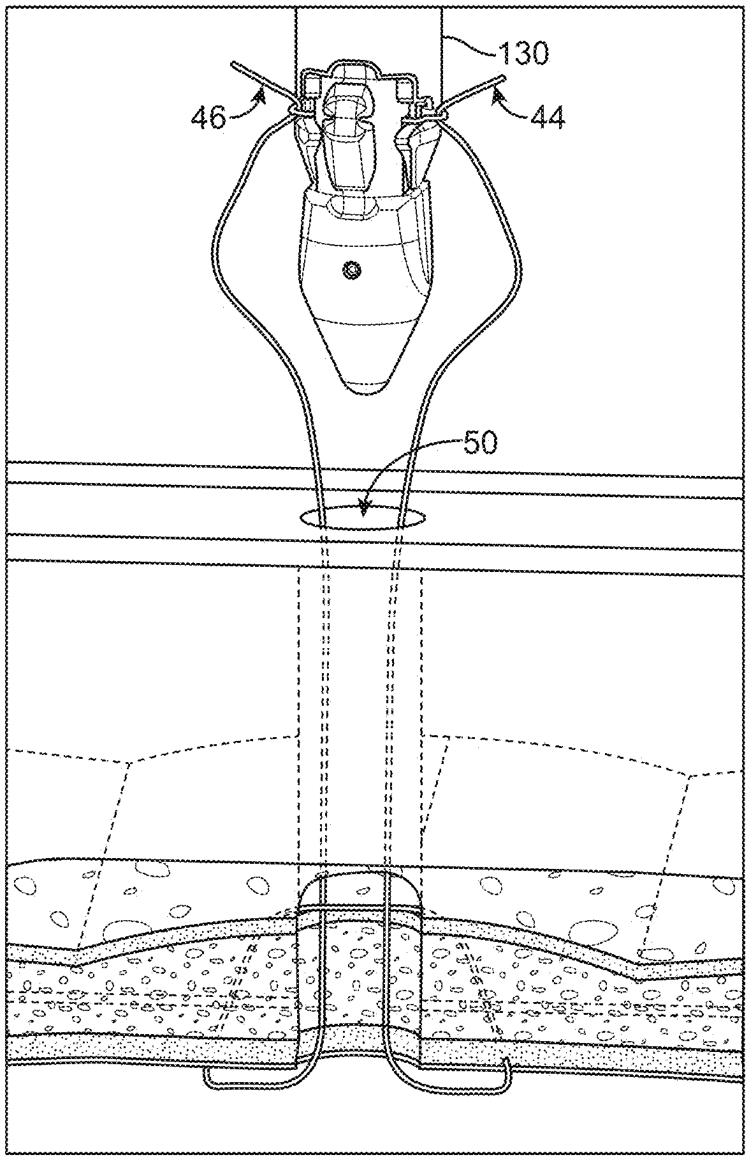
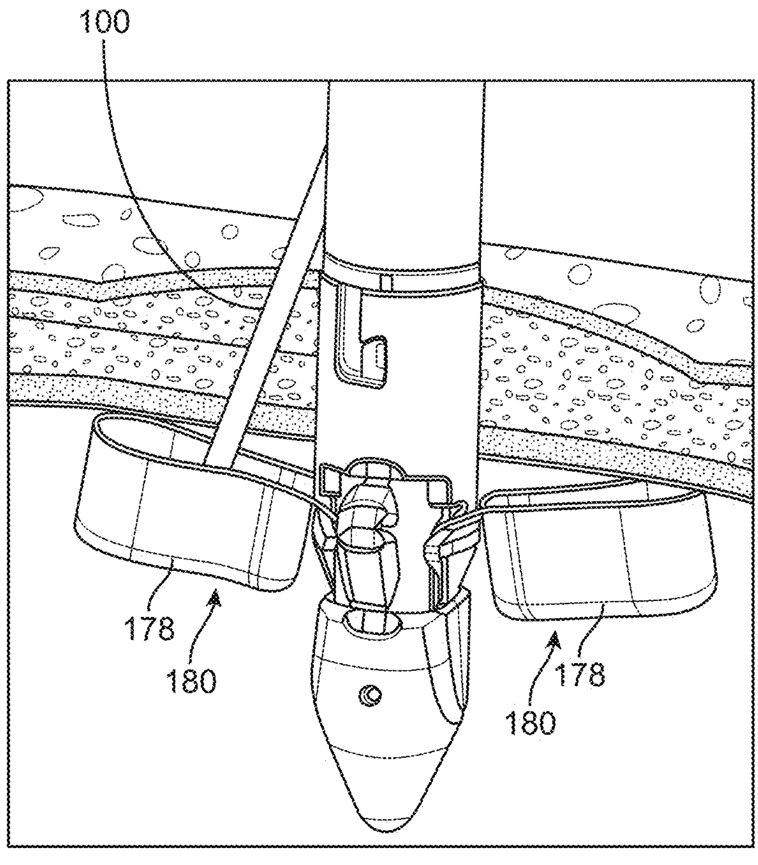


FIG. 13D



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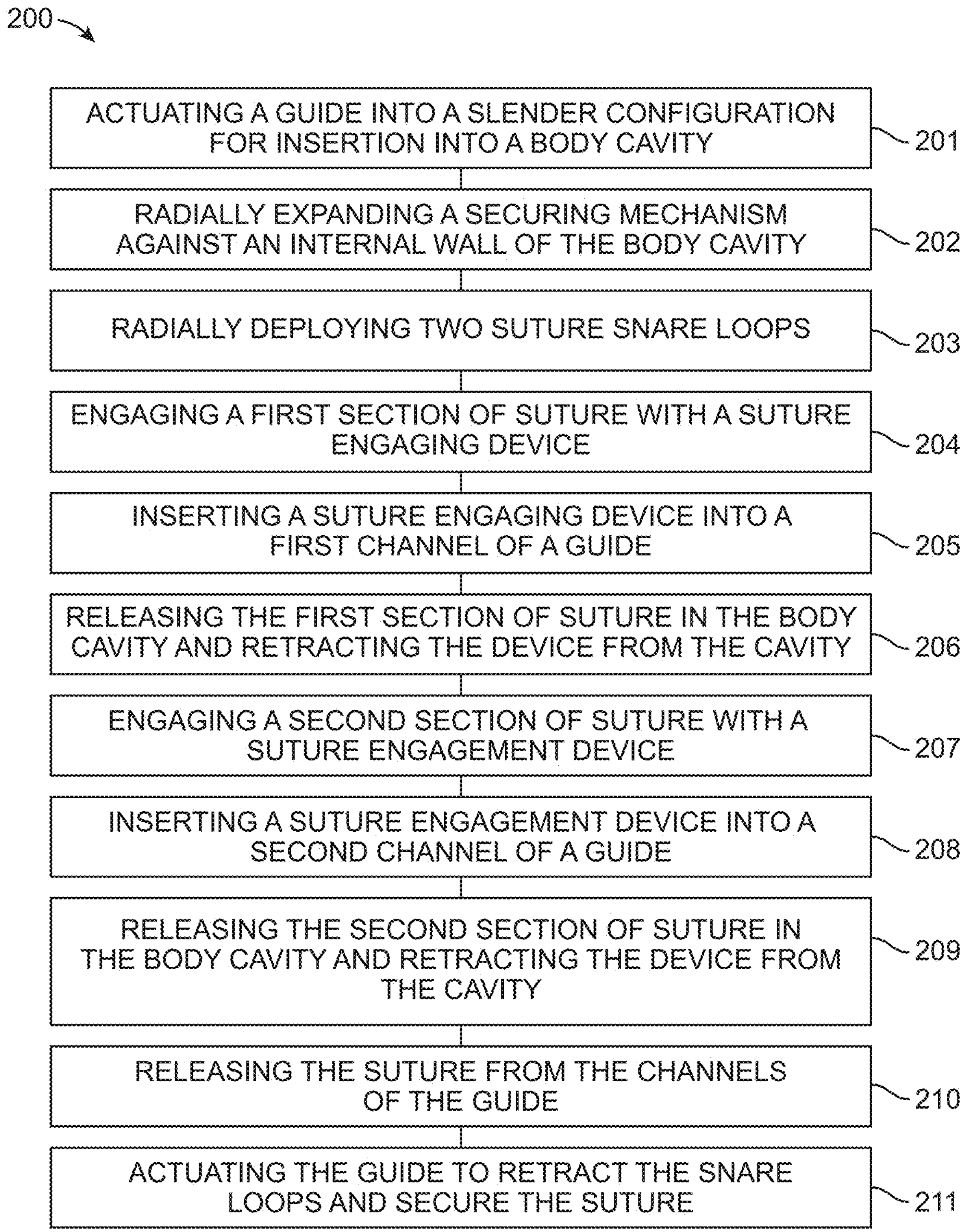

200
ACTUATING A GUIDE INTO A SLENDER CONFIGURATION FOR INSERTION INTO A BODY CAVITY
201
RADIALLY EXPANDING A SECURING MECHANISM AGAINST AN INTERNAL WALL OF THE BODY CAVITY
202
RADIALLY DEPLOYING TWO SUTURE SNARE LOOPS
203
ENGAGING A FIRST SECTION OF SUTURE WITH A SUTURE ENGAGING DEVICE
204
INSERTING A SUTURE ENGAGING DEVICE INTO A FIRST CHANNEL OF A GUIDE
205
RELEASING THE FIRST SECTION OF SUTURE IN THE BODY CAVITY AND RETRACTING THE DEVICE FROM THE CAVITY
206
ENGAGING A SECOND SECTION OF SUTURE WITH A SUTURE ENGAGEMENT DEVICE
207
INSERTING A SUTURE ENGAGEMENT DEVICE INTO A SECOND CHANNEL OF A GUIDE
208
RELEASING THE SECOND SECTION OF SUTURE IN THE BODY CAVITY AND RETRACTING THE DEVICE FROM THE CAVITY
209
RELEASING THE SUTURE FROM THE CHANNELS OF THE GUIDE
210
ACTUATING THE GUIDE TO RETRACT THE SNARE LOOPS AND SECURE THE SUTURE
211

FIG. 15

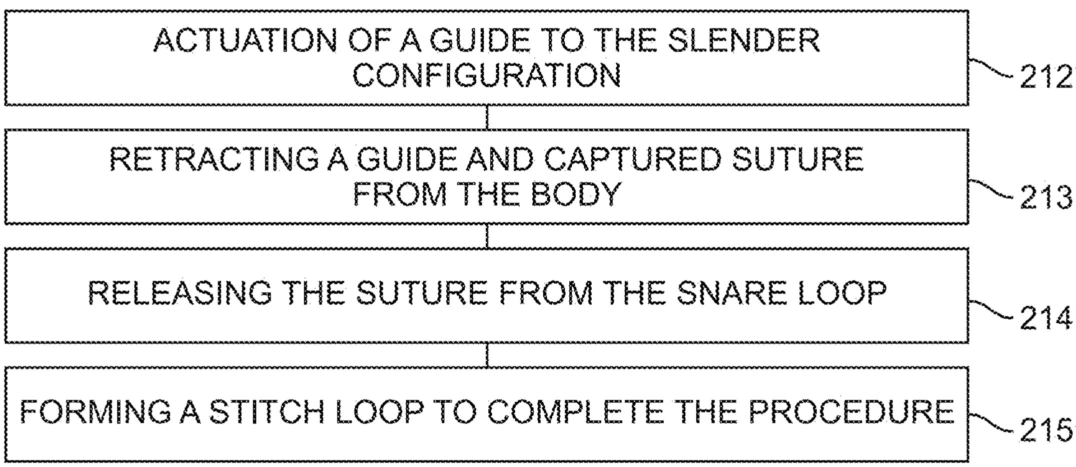

ACTUATION OF A GUIDE TO THE SLENDER CONFIGURATION
212
RETRACTING A GUIDE AND CAPTURED SUTURE FROM THE BODY
213
RELEASING THE SUTURE FROM THE SNARE LOOP
214
FORMING A STITCH LOOP TO COMPLETE THE PROCEDURE
215

FIG. 15 (Cont.)

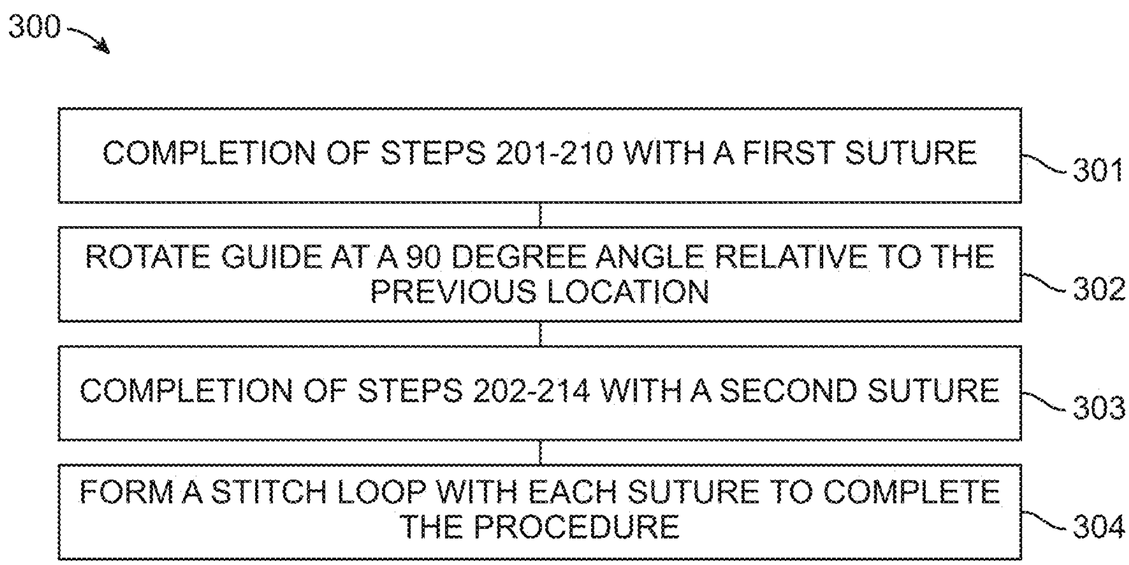

300
COMPLETION OF STEPS 201-210 WITH A FIRST SUTURE
301
ROTATE GUIDE AT A 90 DEGREE ANGLE RELATIVE TO THE PREVIOUS LOCATION
302
COMPLETION OF STEPS 202-214 WITH A SECOND SUTURE
303
FORM A STITCH LOOP WITH EACH SUTURE TO COMPLETE THE PROCEDURE
304

FIG. 16

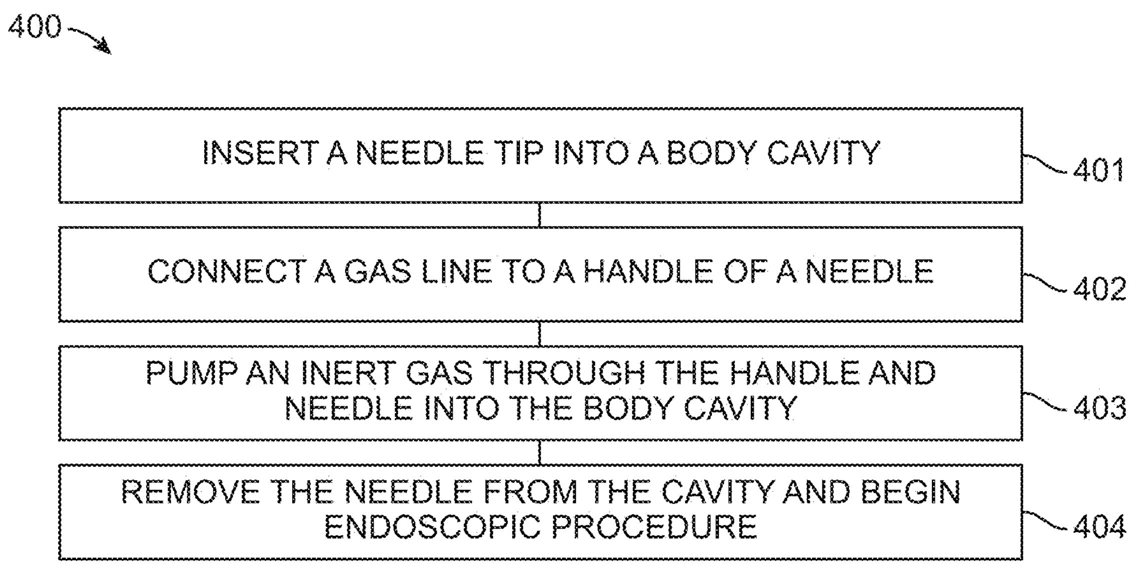
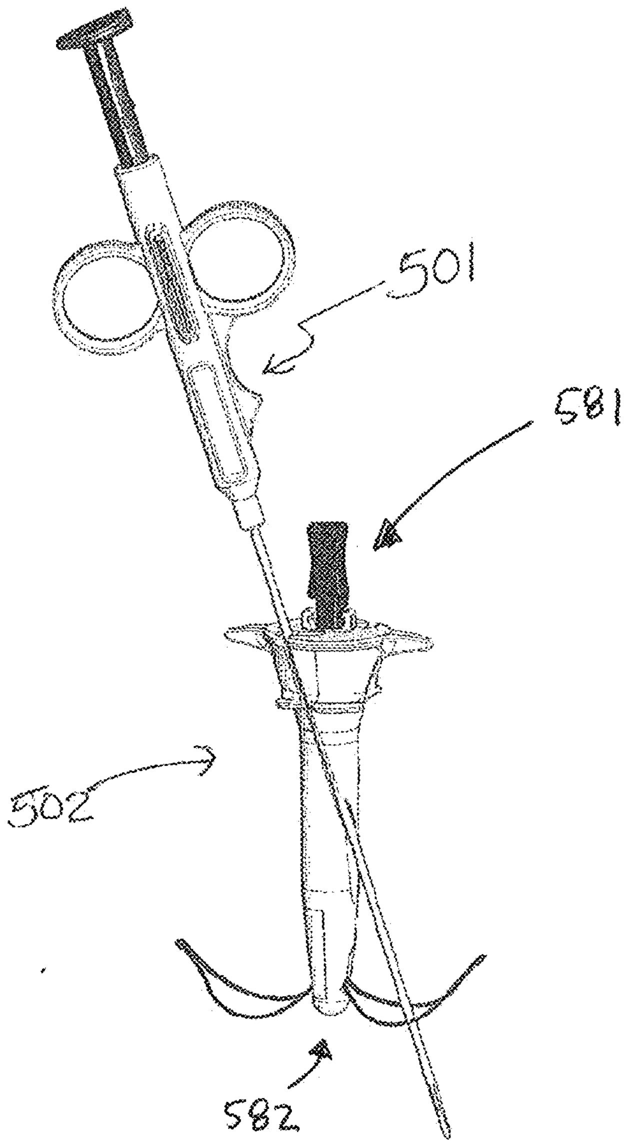

400
INSERT A NEEDLE TIP INTO A BODY CAVITY
401
CONNECT A GAS LINE TO A HANDLE OF A NEEDLE
402
PUMP AN INERT GAS THROUGH THE HANDLE AND NEEDLE INTO THE BODY CAVITY
403
REMOVE THE NEEDLE FROM THE CAVITY AND BEGIN ENDOSCOPIC PROCEDURE
404

FIG. 17



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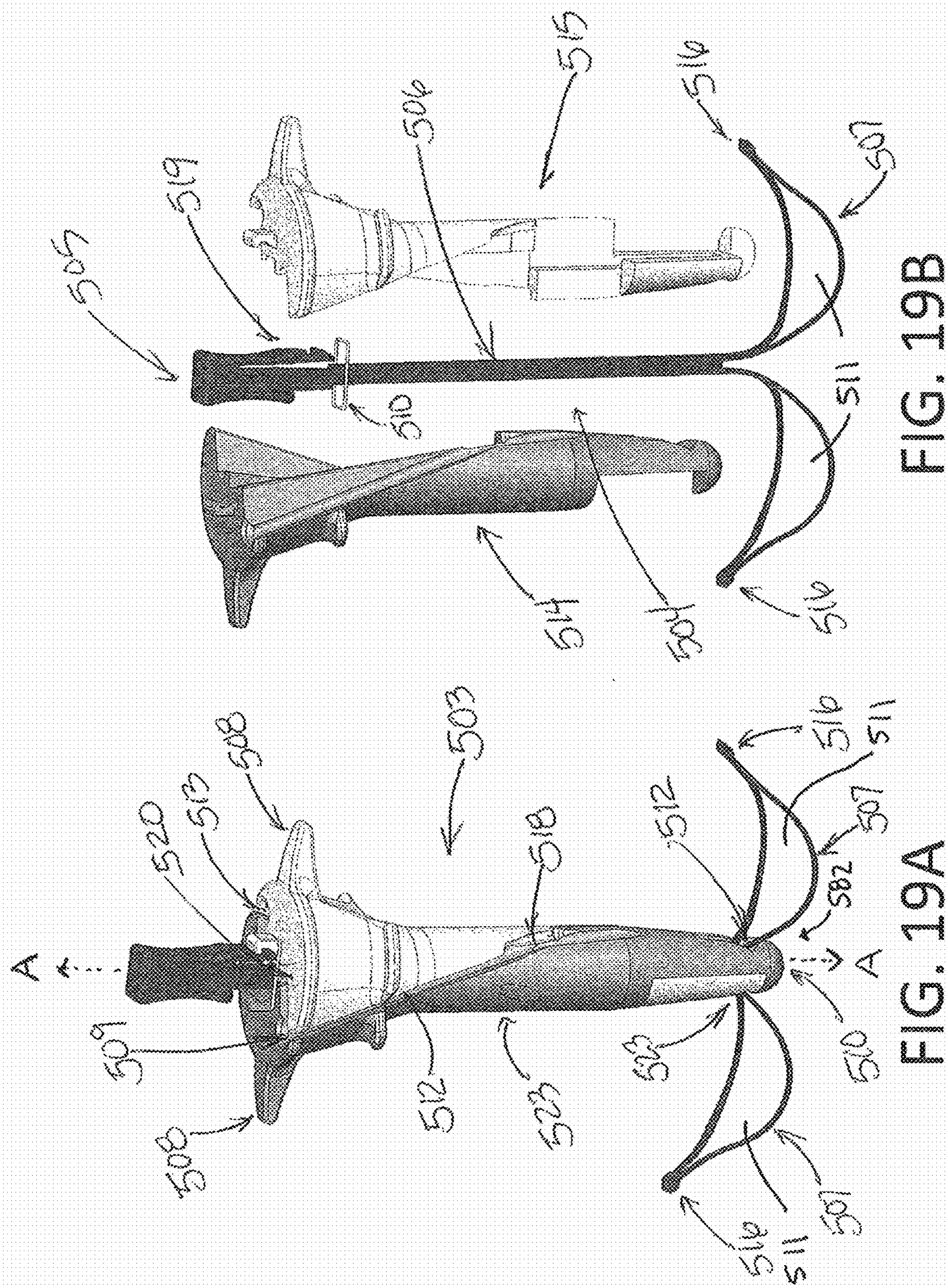

FIG. 19A

FIG. 19B

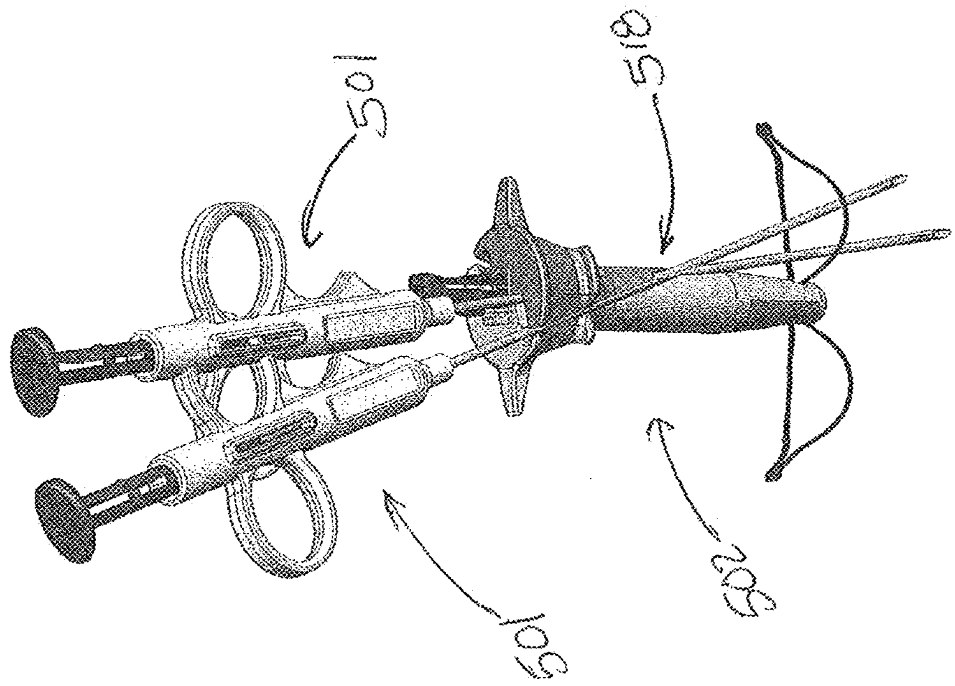

FIG. 20B

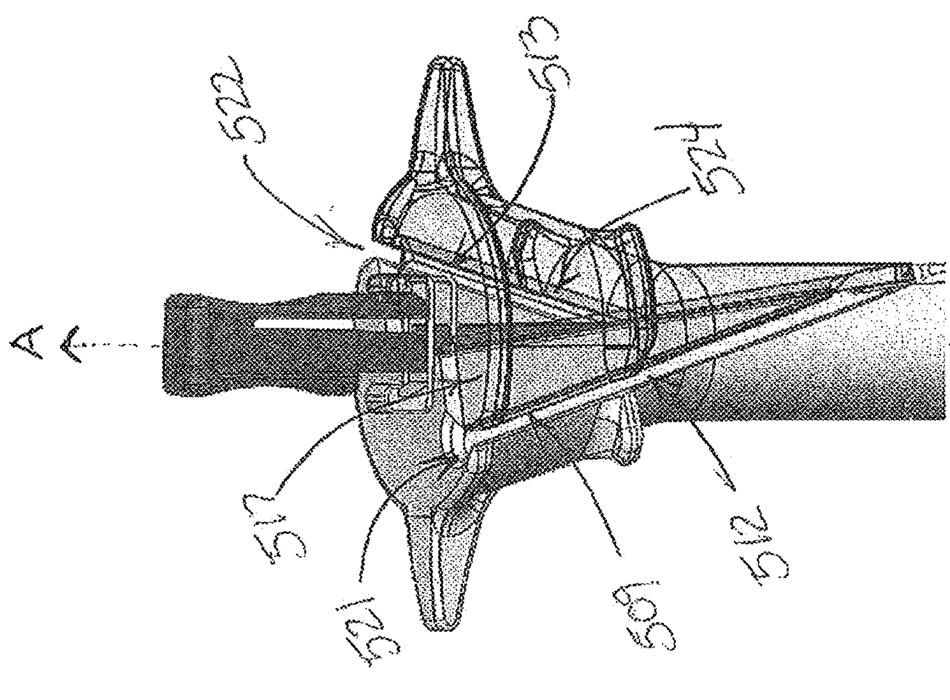

FIG. 20A

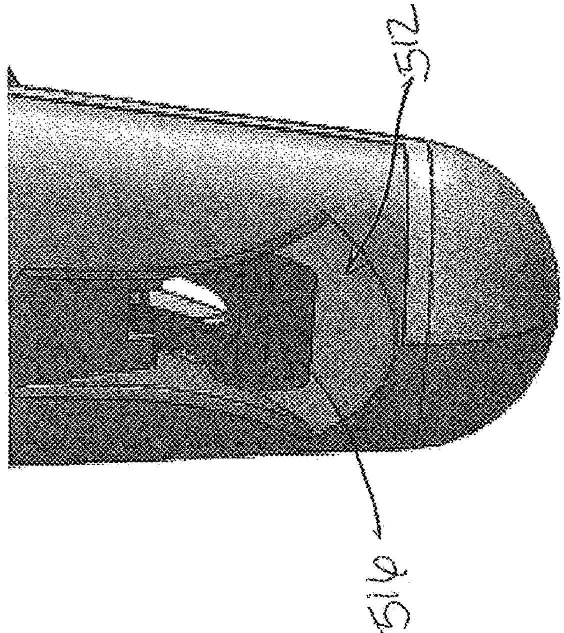

FIG. 21B

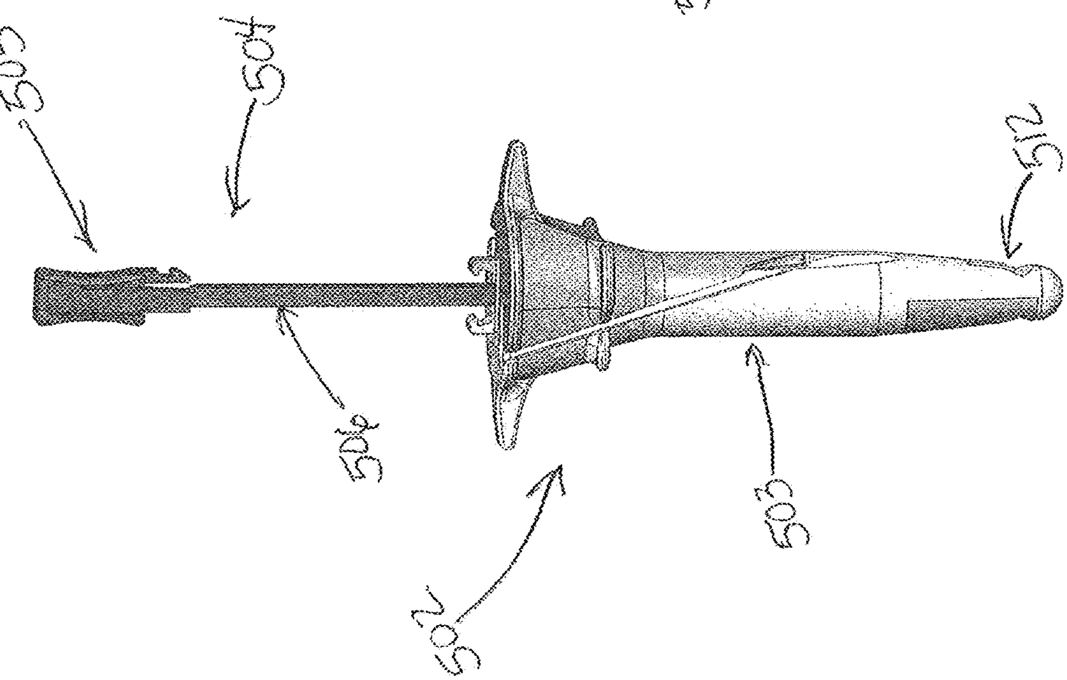

FIG. 21A

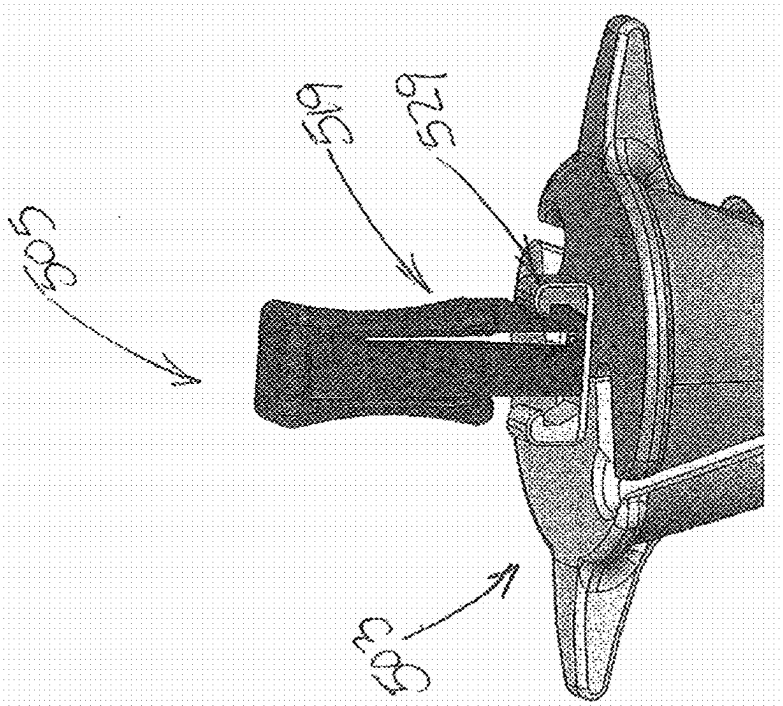

FIG. 23

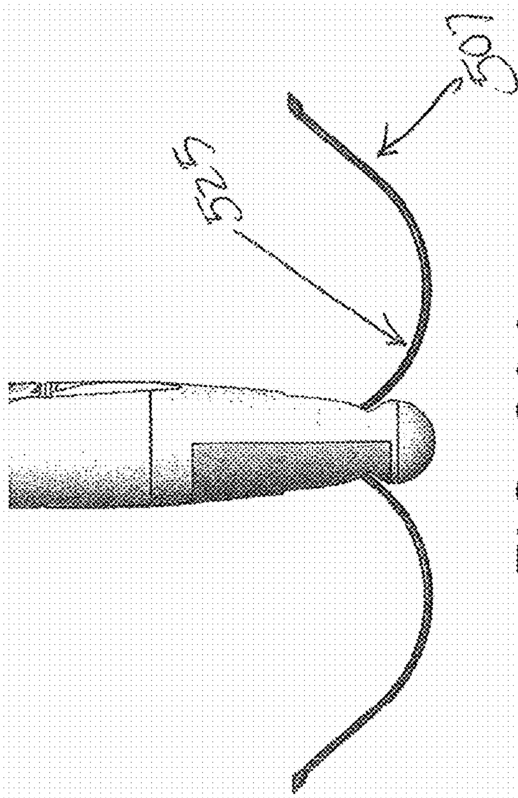

FIG. 22A

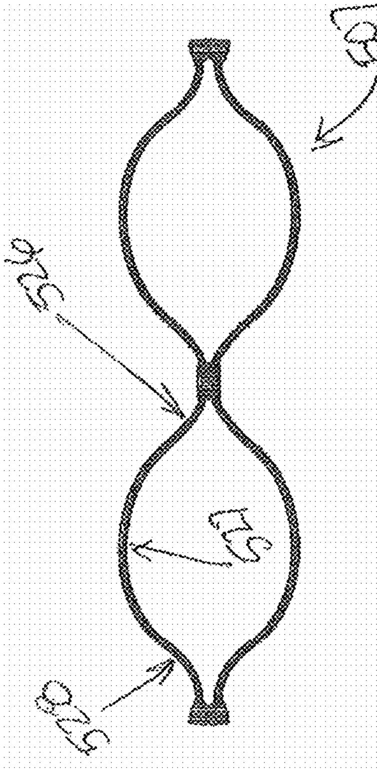

FIG. 22B

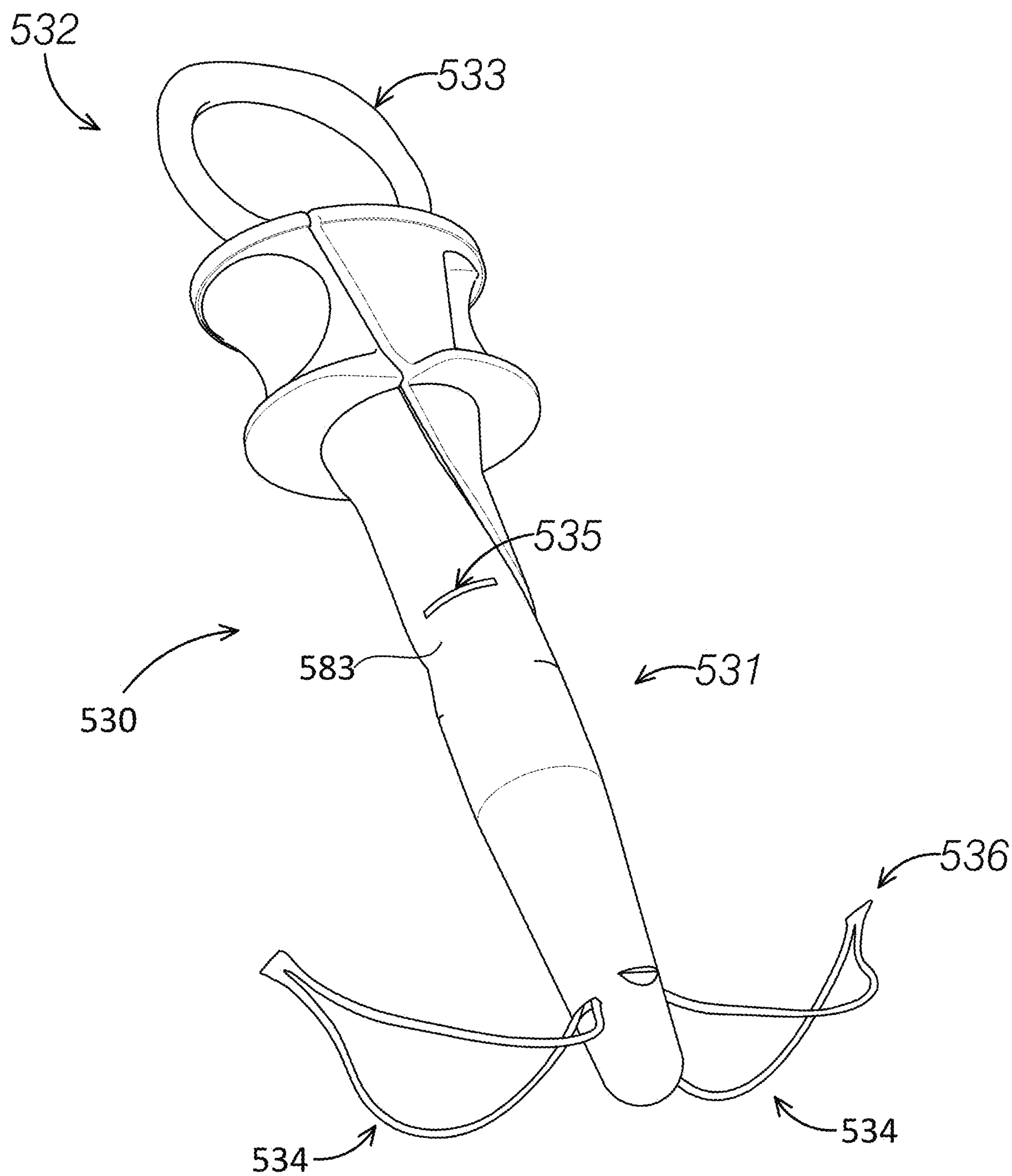

FIG. 24

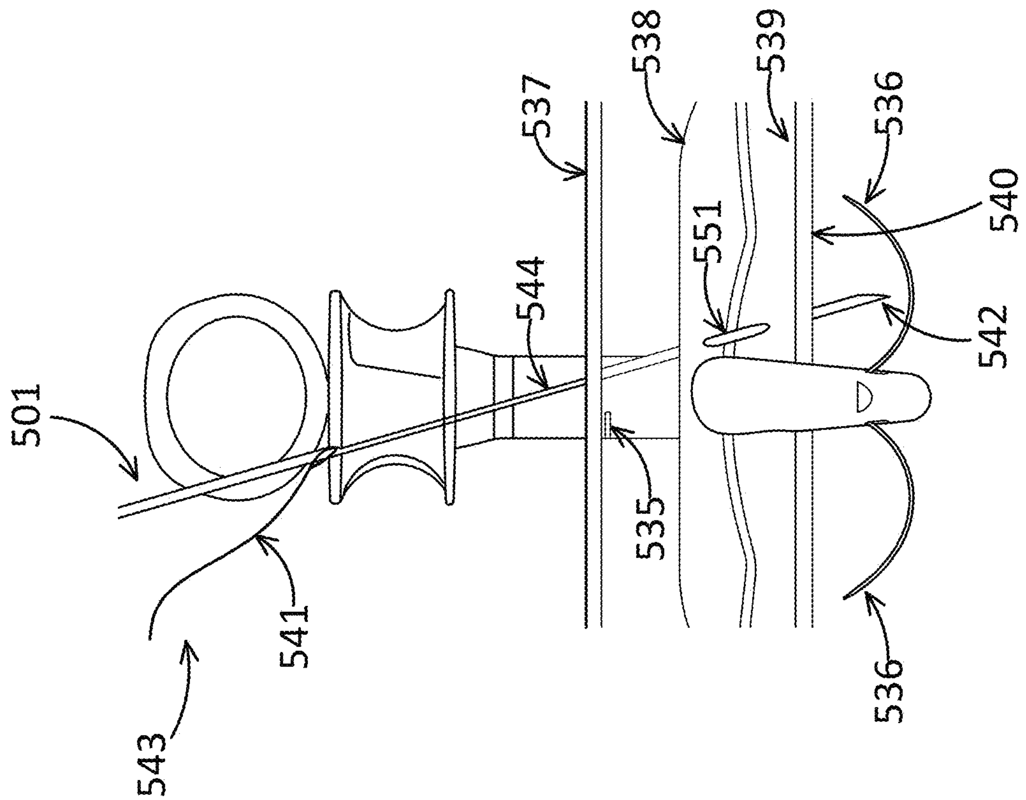

FIG. 25B

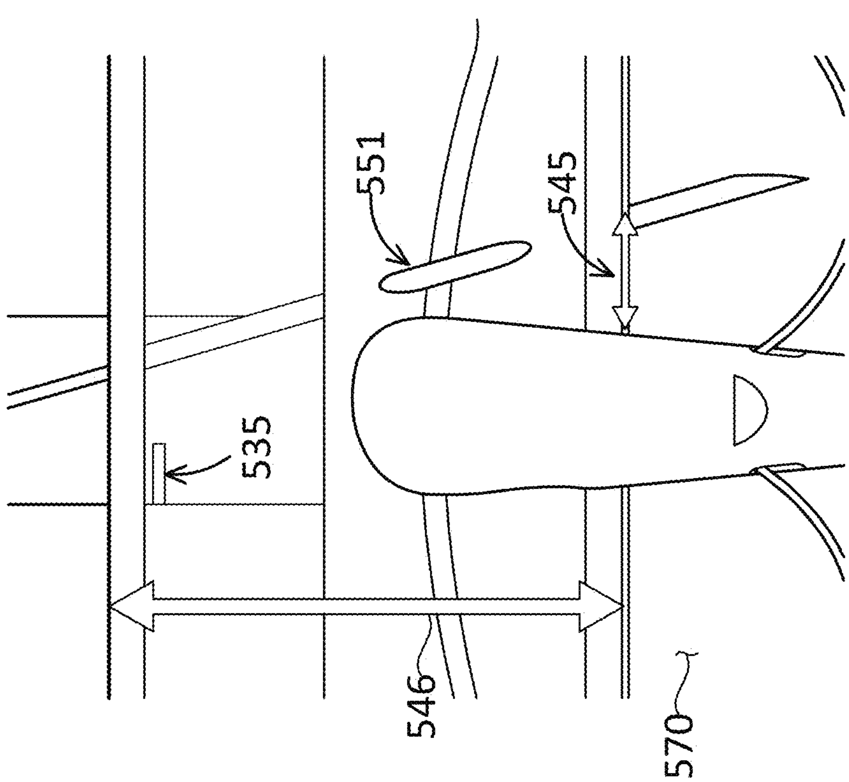

FIG. 25A

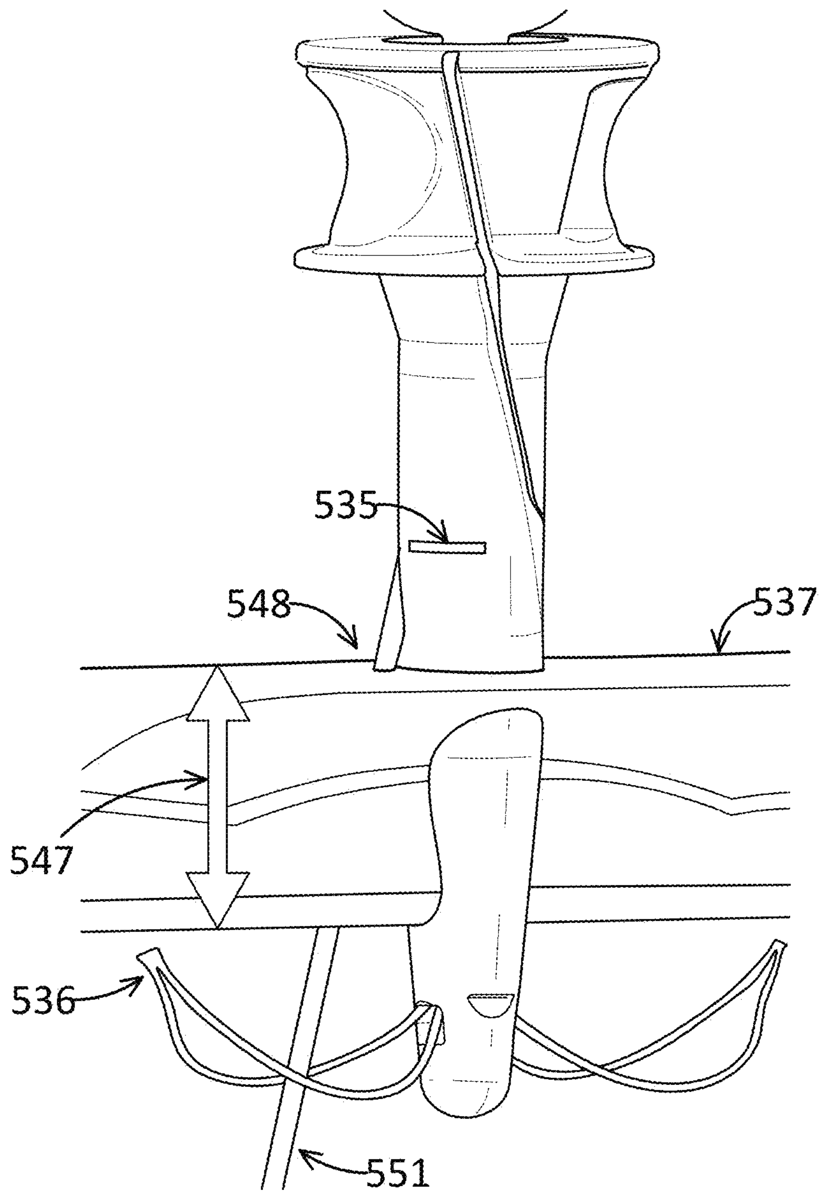

FIG. 26

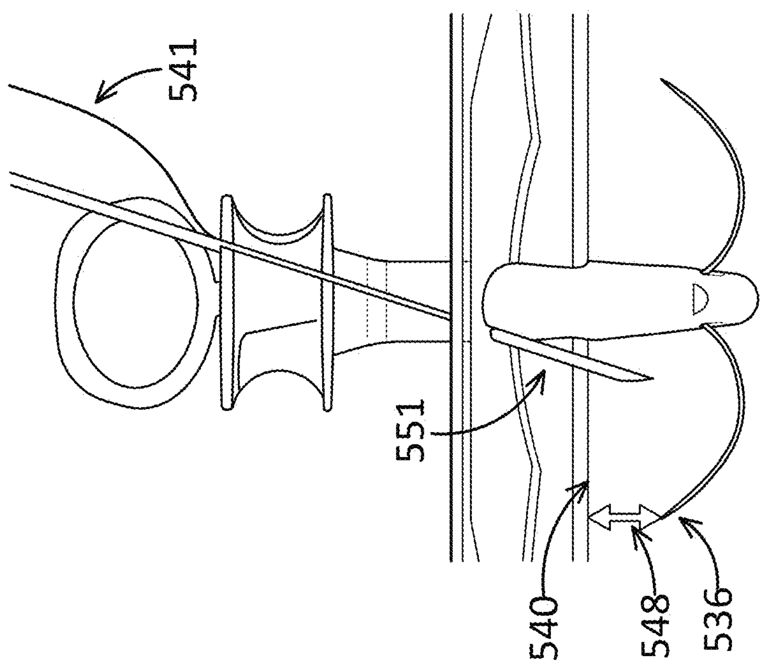

FIG. 27A

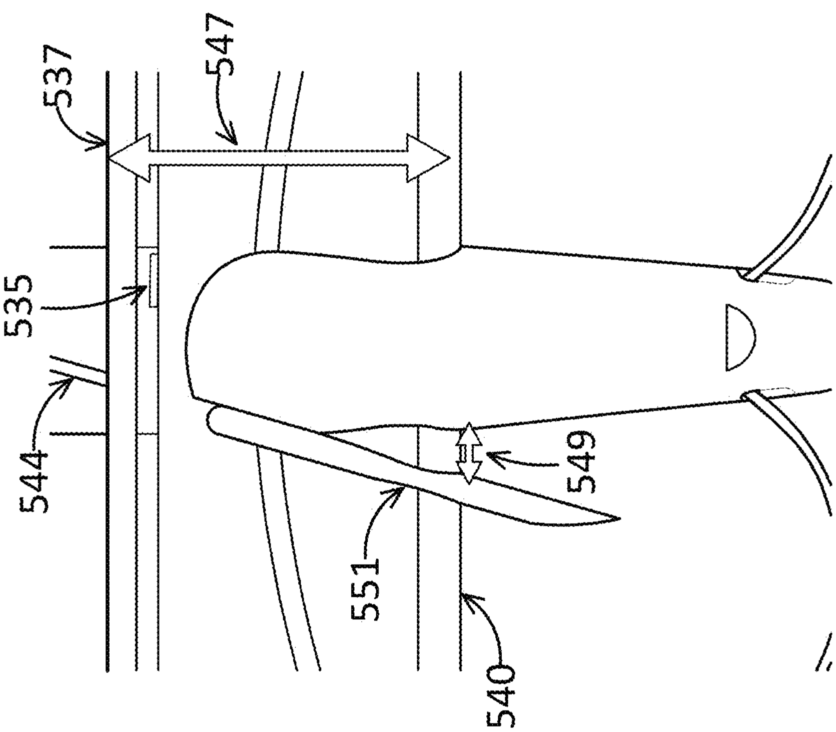

FIG. 27B

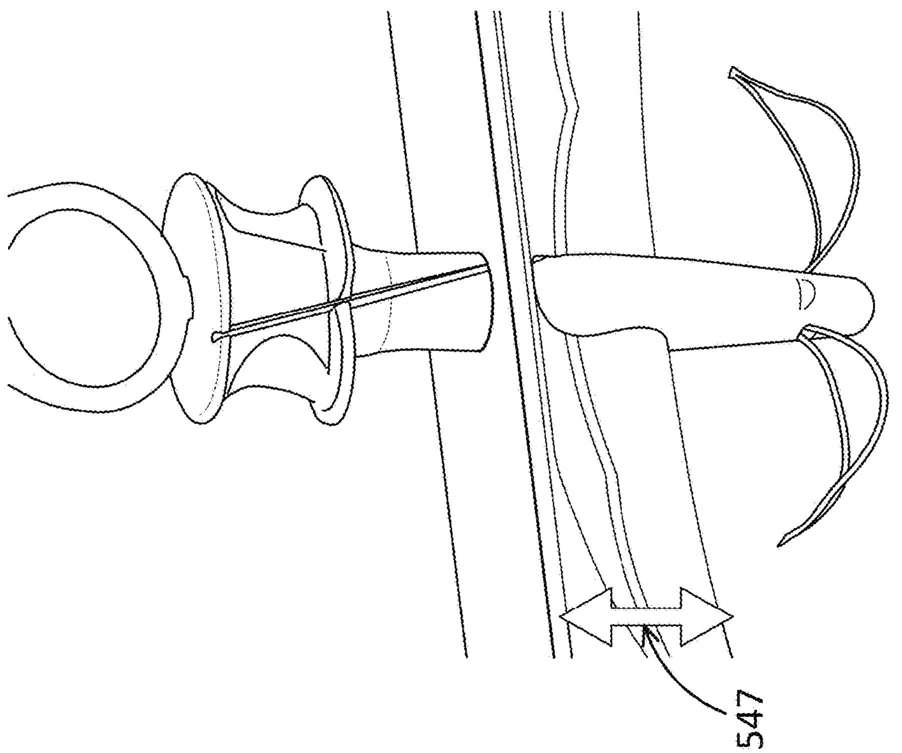

FIG. 28A

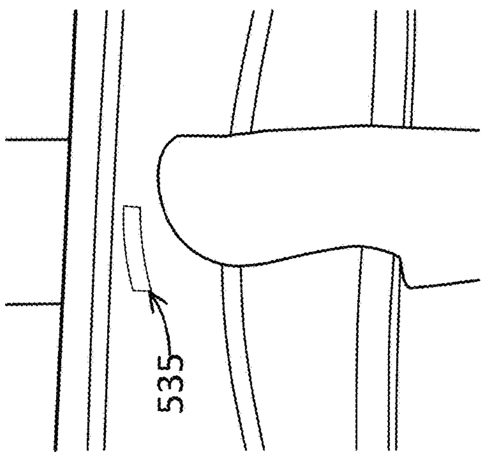

FIG. 28B

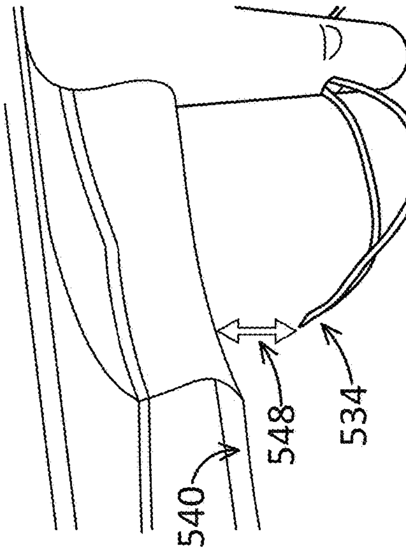

FIG. 28C

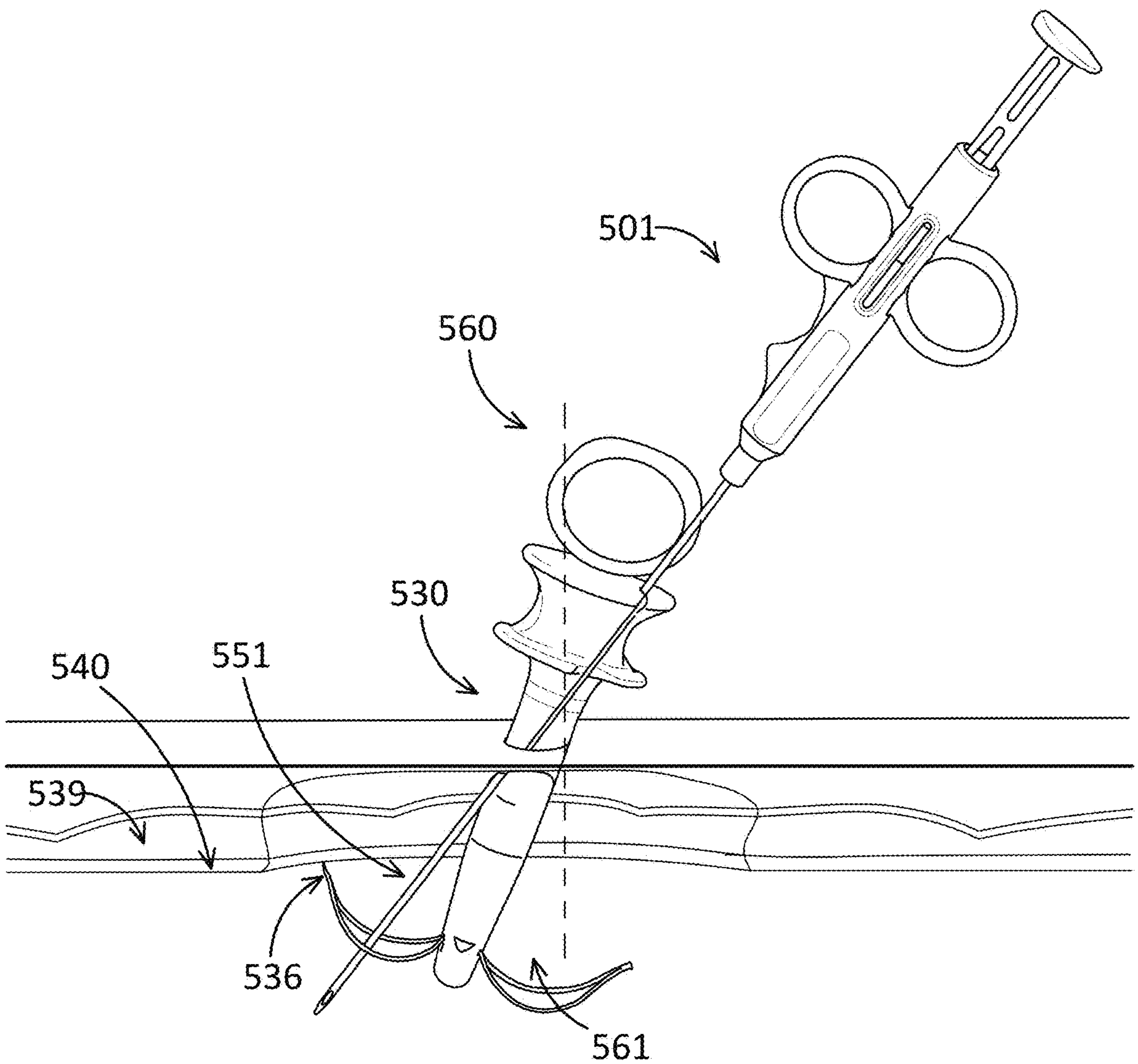

FIG. 29

600

610 ACTUATING A GUIDE INTO A SLENDER CONFIGURATION FOR INSERTION INTO A BODY CAVITY BY RETRACTING A MONOLITHIC SNARE

620 TRANSLATING THE MONOLITHIC SNARE DISTALLY WITH RESPECT TO A BARREL TO DEPLOY AT LEAST TWO SUTURE SNARE LOOPS SO AS TO RADIALLY EXPAND A SECURING MECHANISM AGAINST AN INTERNAL WALL OF THE BODY CAVITY

630 IF A THINNER ABDOMINAL WALL IS INVOLVED, THEN TILTING THE GUIDE IN A FIRST DIRECTION TO CAUSE A DISTAL TIP OF A FIRST SNARE LOOP TO APPROACH OR ABUT AN INNER PERITONEAL WALL

640 ENGAGING A FIRST SECTION OF SUTURE WITH A SUTURE ENGAGING DEVICE, INSERTING THE SUTURE ENGAGING DEVICE INTO A FIRST CHANNEL OF A GUIDE, RELEASING THE FIRST SECTION OF SUTURE IN THE BODY CAVITY AND RETRACTING THE DEVICE FROM THE FIRST CHANNEL

650 IF A THINNER ABDOMINAL WALL IS INVOLVED, THEN TILTING THE GUIDE IN A SECOND DIRECTION OPPOSITE THE FIRST DIRECTION TO CAUSE A DISTAL TIP OF A SECOND SNARE LOOP TO APPROACH OR ABUT THE INNER PERITONEAL WALL

660 ENGAGING A SECOND SECTION OF SUTURE WITH THE SUTURE ENGAGING DEVICE, INSERTING THE SUTURE ENGAGING DEVICE INTO A SECOND CHANNEL OF THE GUIDE, RELEASING THE SECOND SECTION OF SUTURE IN THE BODY CAVITY AT AN EQUIANGULAR SPACING FROM THE RELEASED FIRST SECTION OF SUTURE, AND RETRACTING THE DEVICE FROM THE SECOND CHANNEL

670 ACTUATING THE MONOLITHIC SNARE TO RETRACT THE SNARE LOOPS AND SECURE THE FIRST AND SECOND SECTIONS OF SUTURE WHILE FORMING A SLENDER CONFIGURATION OF THE GUIDE

680 RETRACTING THE GUIDE IN THE SLENDER CONFIGURATION WITH THE CAPTURED SUTURE FROM THE BODY

FIG. 30

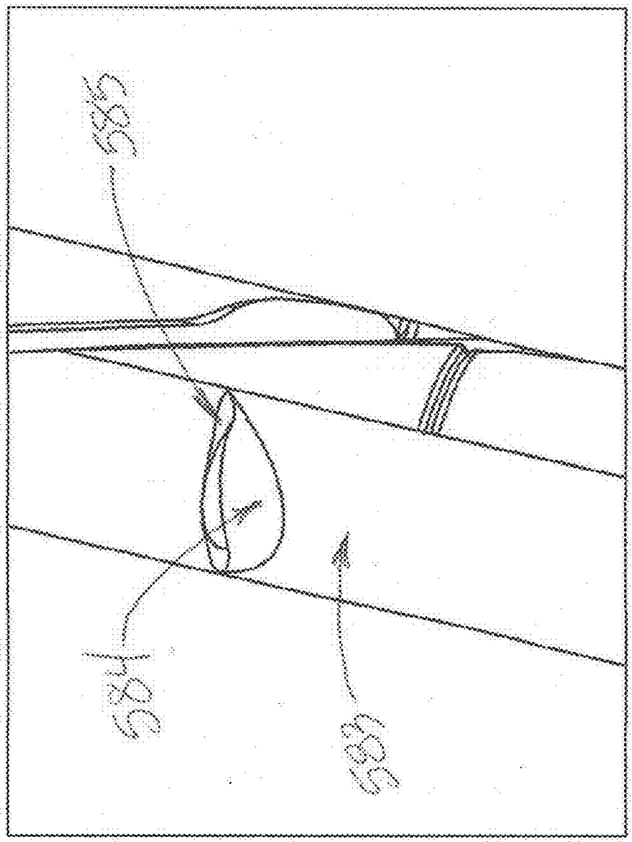

FIG. 32

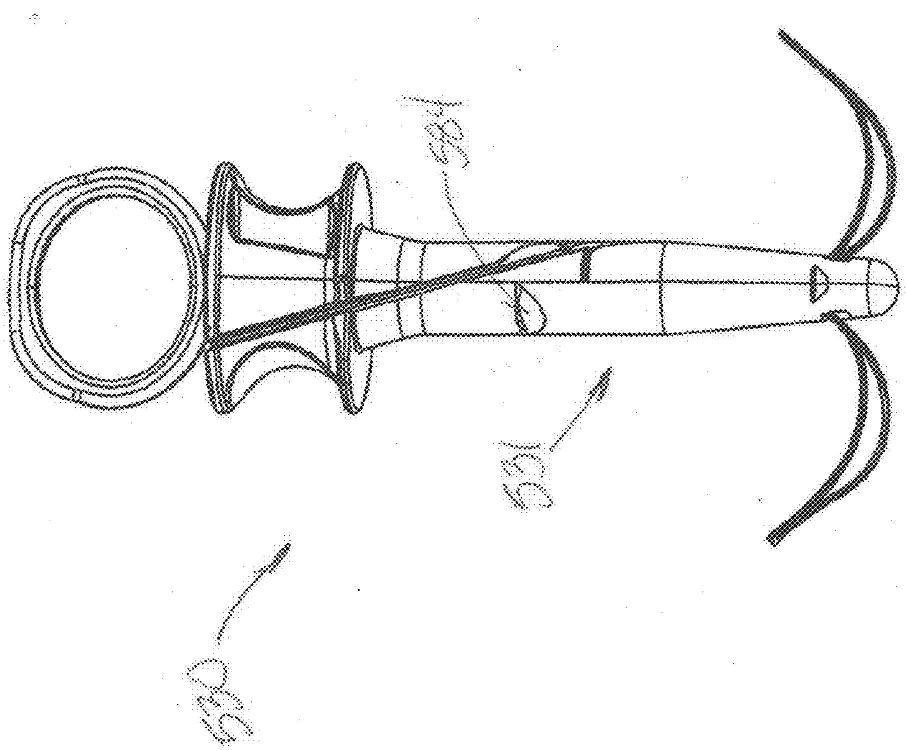

FIG. 31

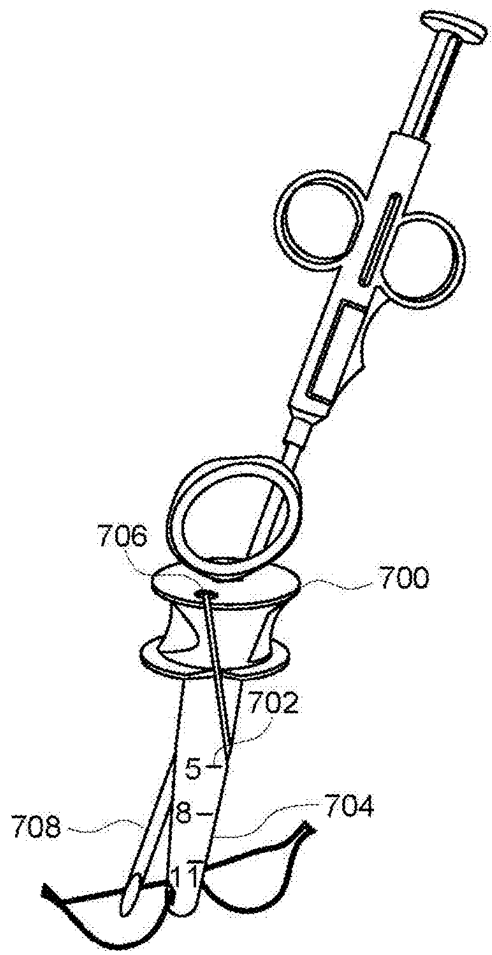

FIG. 33A

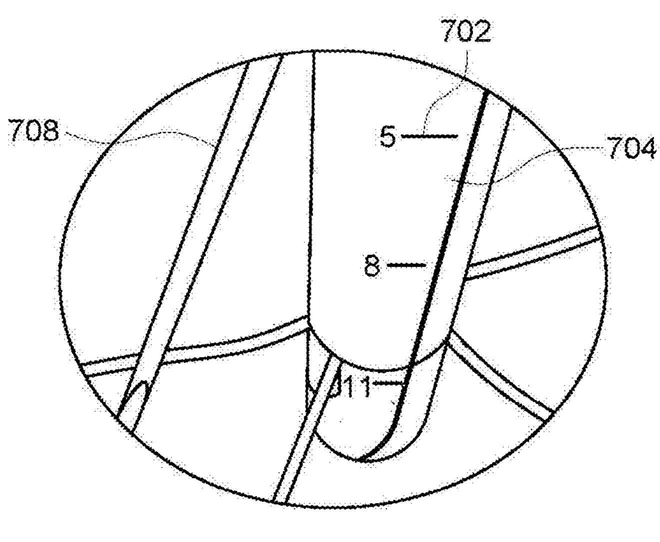

FIG. 33B

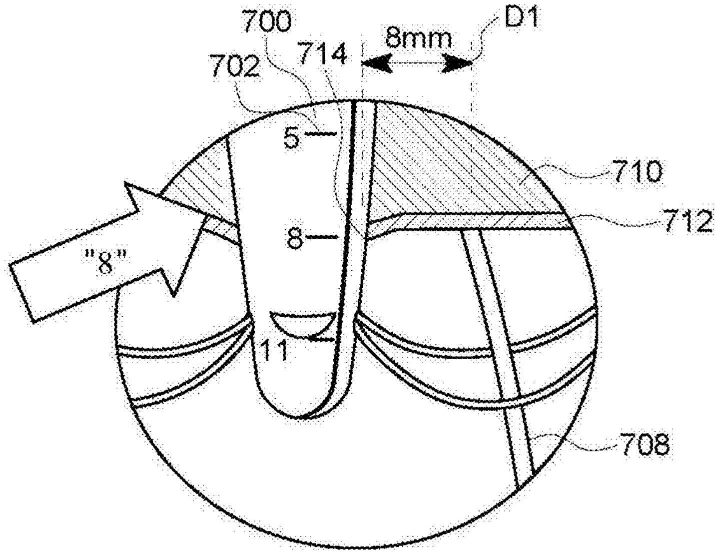

FIG. 34A

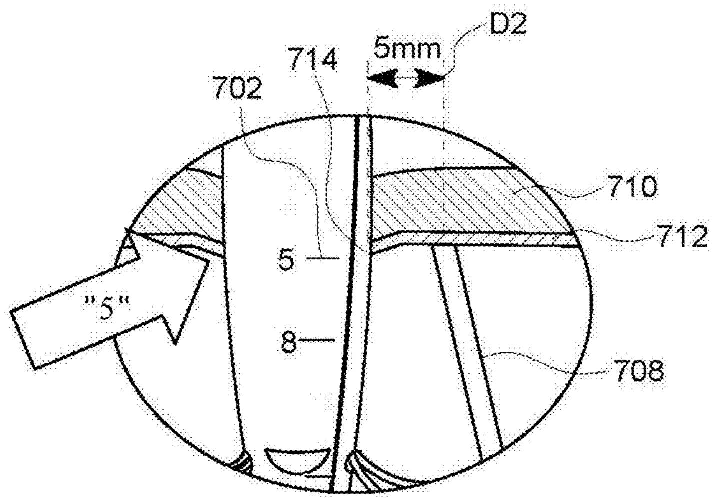

FIG. 34B

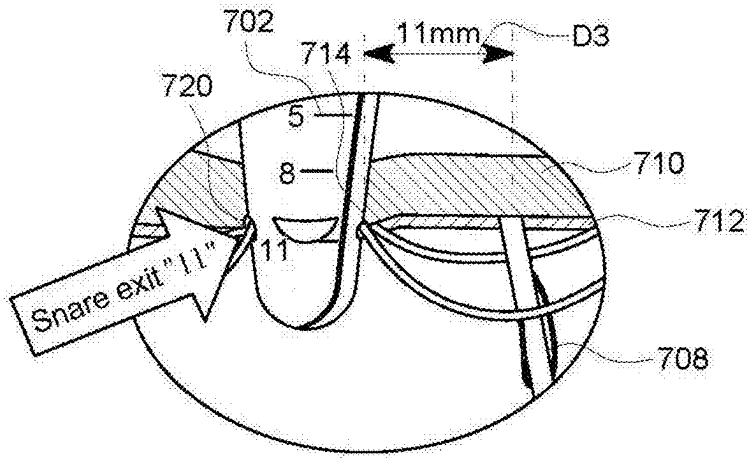

FIG. 34C

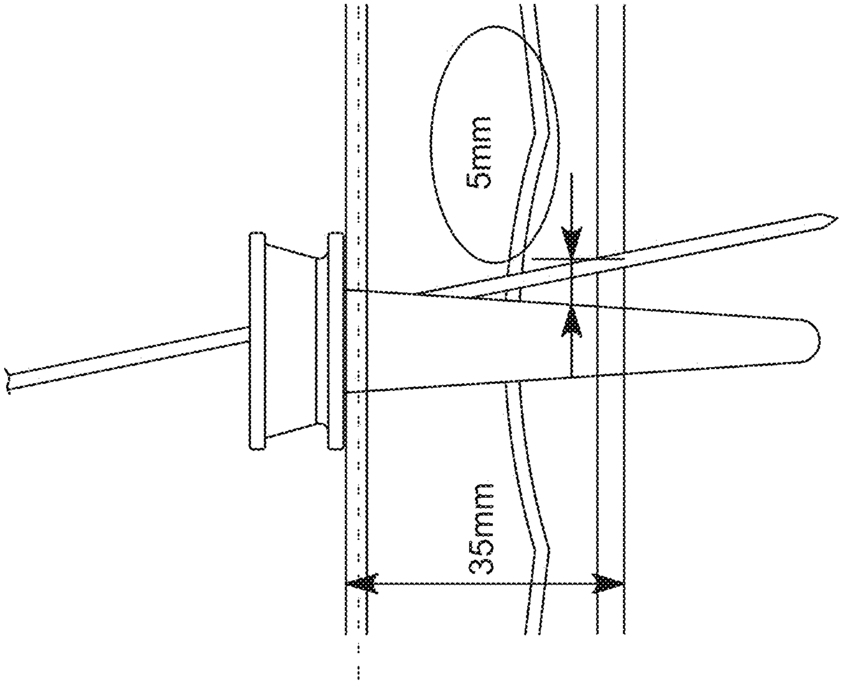

FIG. 35B

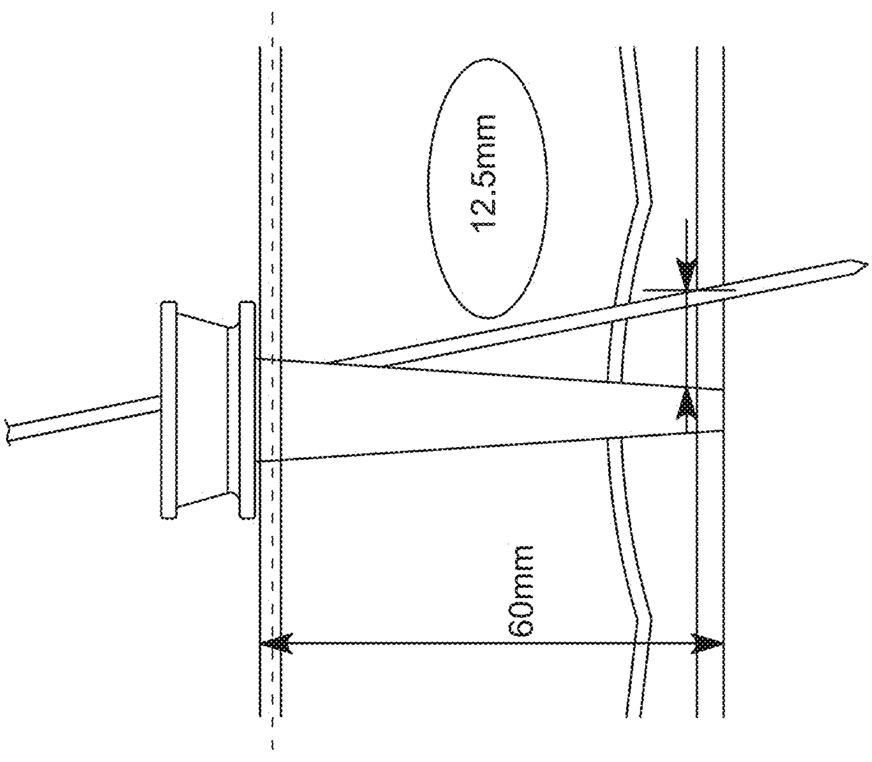

FIG. 35A

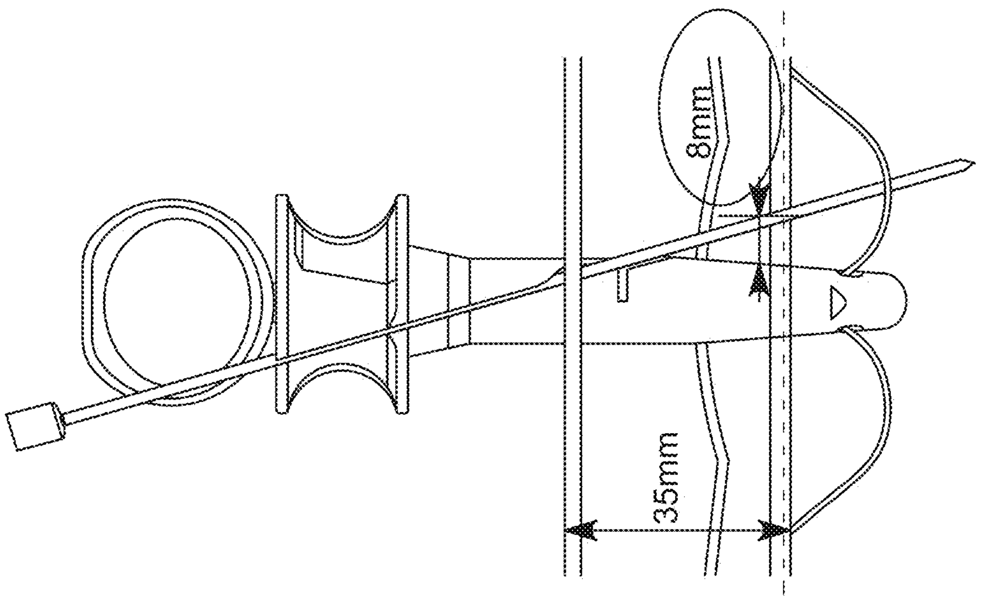

FIG. 36B

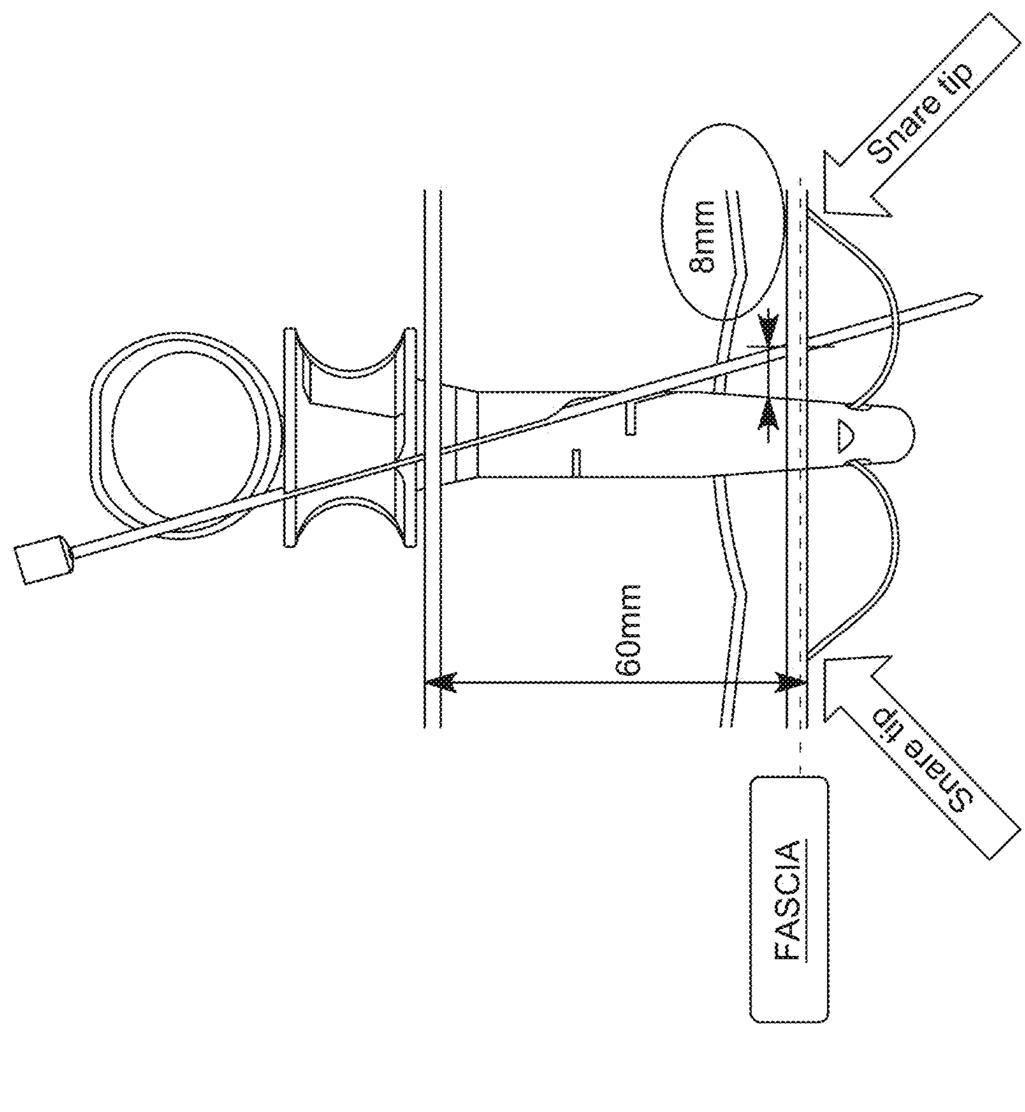

FIG. 36A

METHOD FOR ADJUSTABLY CLOSING TISSUE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of priority of U.S. provisional patent application No. 63/560,641, filed Mar. 2, 2024, the contents of which are herein incorporated by reference

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0002] Embodiments of the invention relate generally to surgical instruments for approximation, ligation and fixation of tissue using a suture, and particularly to the approximation of tissue separated by means of an endosurgical trocar being inserted into a body cavity.

2. Description of Prior Art and Related Information

[0003] The following background information may present examples of specific aspects of the prior art (e.g., without limitation, approaches, facts, or common wisdom) that, while expected to be helpful to further educate the reader as to additional aspects of the prior art, is not to be construed as limiting the present invention, or any embodiments thereof, to anything stated or implied therein or inferred thereupon.

[0004] Numerous methods currently exist for performing laparoscopic procedures. One of the more commonly used methods is known as closed laparoscopy which utilizes a sharp needle (e.g., Veress needle) to puncture the abdominal wall and insufflate the abdominal cavity with an inert gas, such as carbon dioxide, through the needle. This process of insufflating the cavity separates the abdominal wall from the underlying organs creating a gap for the surgeon to work within. A trocar/cannula system is then used to maintain the insufflated cavity and provide a working portal for which instruments can be passed into and out of the abdominal cavity to perform various surgical procedures. When the procedure is completed, it is desirable for the surgeon to close the incision site using suture material to minimize the risk of adverse post-operative events.

[0005] One of the post-operative complications associated with this procedure is the incidence of trocar site hernias, where a portion of an organ or fatty tissue protrudes out through the hole in the abdominal wall created by the trocar access portal. It is believed that improper closure, or complete lack of closure, of the incision site at the peritoneum is the primary cause of these hernias which form during the post-operative period ranging from several days to several months following the procedure. Traditional methods of wound site closure require an additional set of instruments (suture passers, guides, etc.) to be introduced into the surgery. A number of these instruments have been previously disclosed. However, the prior art related to trocar wound site closure instrumentation are typically cumbersome to use and do not provide for a simple, reproducible, and reliable means of closing the wound site.

[0006] In view of the foregoing, there is a need for improved methods for trocar would site closure and instruments for performing such methods.

SUMMARY OF THE INVENTION

[0007] An exemplary system according to embodiments of the invention comprises a surgical instrument as well as a surgical instrument set that may have certain functions. First, the system may have the capabilities to provide entry into the abdominal cavity and subsequently insufflating the cavity for use in laparoscopic surgical procedures. In one exemplary embodiment, the system comprises a needle apparatus having a sharp needle tip and an insufflation channel to facilitate penetration into the abdominal cavity and insufflation. A unique obturator tip is provided to shield the sharp needle tip upon insertion into the cavity. Second, the system may have the capabilities to close the fascial/peritoneal layer at the trocar wound site in a quick, consistent and reproducible manner at the end of the procedure. To facilitate closure of the wound, the system includes the same needle used in combination with a guide apparatus which has suture capture features disposed at or near the distal tip.

[0008] The needle apparatus may also serve as a suture passer, in that it has the ability to carry and retrieve suture through tissue layers for suturing closed the wound site. The needle apparatus also has the ability to insufflate the abdomen during the laparoscopic procedure. The needle apparatus may comprise several components including: a handle, actuation mechanism, a connector for connecting the needle to a gas line, a capture rod, an outer needle shaft, and a spring loaded safety tip on a hollow obturator tube.

[0009] In an exemplary embodiment, a handle at the proximal end of the needle apparatus allows for single-handed or double-handed use. The handle may also contain a finger loop or loops for additional security while holding the needle. An actuator mechanism may be disposed adjacent to the handle and configured for the deployment and retraction of the capture rod used to secure the suture material within the tip of the needle. The actuator mechanism may include a sliding plunger that translates along the long axis of the handle that moves the capture rod between a first position in an axially extended configuration and a second position in an axially retracted configuration. The actuator may be spring loaded in one direction such that the capture rod is biased to the retracted position. This may allow the suture to be passively captured without actuation of the plunger. The handle and actuator means may be constructed from metals (such as stainless steel, titanium or aluminum) or plastics (such as polyacetal, nylon, polypropylene, poly-ether-ether-ketone, or polycarbonate), or any combination of the two.

[0010] A long outer needle shaft may be connected to the proximal handle and extends distally over a length that may range from 2-38 centimeters, or more typically between 10-20 centimeters. The outer needle shaft may have a sharp tip, or needle peak, at the distal-most point to ease the insertion of the needle through the various tissue layers. The outer shaft may house an obturator tube that has a hollow, unobstructed inner lumen, with a blunt tip. The obturator tube may also house a capture rod used for securing the suture for passing through tissue. The outer needle shaft, obturator tube and capture rod would optimally be constructed from metals such as stainless steel, titanium or aluminum.

[0011] The distal-most end of the obturator tube may have a blunt or rounded plug or surface at the tip. The entire obturator tube may be spring loaded to allow for the blunt tip to translate away from the tip of the needle when it is loaded, and passively travel back to the tip of the needle when it is unloaded. The obturator spring may be housed within the handle. The spring loaded obturator would serve as a safety mechanism for protecting the internal organs within the abdomen after the needle is passed through the abdominal wall.

[0012] A portion of the wall of the outer needle shaft may be cutout near the distal tip which may be used to create a slot to accommodate the suture during the suture passing process. Similarly, a portion of the wall of the obturator tube may be cut out near the distal tip of the tube to provide an opening for the capture rod to secure the suture to the wall of the outer needle shaft. The window cutout in the obturator tube must be long enough such that it can accommodate the suture as it travels back and forth. Lastly, the capture rod has a slot with one or more ramped faces. A distal ramped surface on the capture rod slot is used to capture the suture against the outer needle shaft. A proximal ramped surface may assist in pushing the suture out of the window in the obturator tube, facilitating the release of the suture from the needle.

[0013] The needle capture rod is used to secure the suture to the needle for suture passing activities. Initially the actuator may be pressed to extend the capture rod and expose the slot in the capture rod. A section of suture may be placed into the slot, and the actuator is released to retract the capture rod. As the capture rod retracts, the suture becomes trapped between the distal surface of the slot in the capture rod and the cutout in the outer needle shaft. When the suture needs to be released, the actuator may be pressed again to extend the capture rod. As the capture rod is extended the proximal face of the slot may push the suture material out of the cutout in the obturator tube and away from the needle shaft.

[0014] In another exemplary embodiment, a luer connector or other quick connect type device may be disposed on the proximal handle to provide an entry passageway for the gas to enter into the needle. The unobstructed inner lumen of the obturator tube may allow for the passage of an inert gas for insufflation of the abdomen.

[0015] The guide apparatus may serve dual purposes, as it first may be used to guide the needle through the abdominal wall in a repeatable manner, and second used to capture the suture material after it is passed into the abdominal wall. The guide may comprise a slotted barrel, collapsible barrel tip, plunger, main shaft, cap, suture catchers such as a capturing snare cord, and guide tubes along with various fasteners and springs.

[0016] The slotted barrel may have two slotted channels to accommodate the passage of the needle. The entries and exits of the two channels may be spaced 180 degrees radially apart from each other such that the stitch can be placed on opposing sides of the wound. The channels' purpose is to guide the needle repeatably through the same tissue thickness and into the suture snare cord, where the suture can be released. The trajectory of the channels is referenced off the inner wall of the peritoneum such that approximately 5-15 millimeters of tissue bite is achieved from the periphery of the wound. The proximal ends of the channels may have a widened and or tapered opening to ease the entry of the needle into the channels. Slots in the channels will allow for the middle section of the length of suture to be released from the constraints of the guide channels. The width of the slots in the channels should be large enough for the suture to easily be released from the channels, yet small enough to not allow the needle to exit the channel or get caught against it.

[0017] The guide may comprise a main shaft that is slidably disposed within the slotted barrel of the guide. The main shaft may be used to actuate the expanding arms, comprising living hinges in the preferred embodiment on the collapsible barrel tip. One or more expanding arms may be used to locate the guide against the inner peritoneal wall as a reference point to ensure consistent tissue bite depth of the needle, as previously described. The main shaft may be spring loaded in a proximal position such that the expanding arms are biased to a radially expanded position where the outer profile of the arms exceeds the diameter of the slotted barrel. As the main shaft moves distally, the arms may be contracted such that aligns their outer diameter with the outer diameter of the slotted barrel in a continuous slender fashion. The main shaft may be connected at the distal end to the barrel tip, and connected to the plunger on the proximal end.

[0018] The distal end of the barrel tip may have a blunt tip to minimize the potential of harm or damage to the adjacent tissues during insertion. Moving proximally away from the blunt tip, the outer wall of the barrel tip may have a tapered region that gradually radially increases to the outer profile of the guide as designated by the outer diameter of the slotted barrel. The tapered section may facilitate the ease of insertion of the guide into the trocar wound site.

[0019] The barrel tip may have one or more stop tabs that provide a hard stop for the barrel tip as the expanding arms are actuated, to prevent excessive flexion in the hinge material. Along the length of the stop tabs, a cutout section may exist for the suture catcher, such as snare loop, to be retracted into for capturing the suture material against the guide.

[0020] The guide may have a slider that is used to actuate the snare cord material. The slider may be slidably disposed on the slotted barrel. Two suture catchers, such as snare cords, may be connected at their ends to the slider body, with a loop formed at the distal tip of the guide. The slider may be spring loaded such that the snare cords are biased into a radially extended position. As the slider is pulled proximally, the snare cord is retracted against the extension arms of the barrel tip. As the slider is released distally, the snare cord is radially extended out and away from the barrel to create two snare loops for the suture to be passed into. The slider may have two tabs that can be used to pull the slider proximally using one or more fingers on each tab. The snare cords may be constructed from a mono- or multi-filament wire that has the flexibility to easily bend and conform to various geometries yet stiff enough to create a self-supported snare loop that extends generally perpendicular to the long axis of the guide. Materials that may be used to construct the snare cord include plastics such as nylon, polyethylene, polyester or polypropylene or metals such as stainless steel or nitinol.

[0021] A plunger at the proximal end of the guide may be used to provide a counterforce when pulling on the slider. As the plunger is pushed and the slider is simultaneously pulled, the snares will move into the retracted position first, and then the expanding arms are retracted into the slender configuration. As the slider is released, the spring forces will extend the snares to the extended position and the expanding arms will be converted to the radially expanded condition.

[0022] In another embodiment, the snares may include a basket element to prevent the needle from traversing deeply into the abdominal cavity and causing potential harm. The basic procedural steps of the utilization of the suturing system may flow as follows. At the end of the surgical procedure, the trocar is removed from the body exposing the wound. The slider on the guide is pulled up against the plunger to contract the flexing arms and retract the snares such that the profile of the guide is at its minimum. The guide can then be inserted into the wound with the plunger continually pulled against the slider. The slider and plunger are then released expanding the arms and deploying the snares. The guide can be pulled upward and away from the body cavity until the arms rest against the inner wall of the peritoneum. A short tail at one end of the suture is secured by the capture rod in the tip of the needle. The needle, with suture, is then passed through a first needle channel in the guide and is advanced through the guide, tissue and snare, into the abdominal cavity. The needle then releases the suture into the cavity, and is retracted from the body. A second short tail at the second free end of the suture is then secured by the capture rod in the needle. The needle, with suture, is then passed through the second needle channel in the guide and advanced through the guide, tissue and snare, into the abdominal cavity. The needle then releases the suture into the abdominal cavity and removed from the guide and body. The remaining loop of suture outside the body may then be released from each of the slots in the needle tracks. The slider on the guide is then again pulled against the plunger to retract the snares, capturing the free ends of suture, and contract the flexing arms allowing the guide to be removed from the wound, carrying the suture with it. Once outside the body, the snares may need to be deployed enough to release the free ends of the suture. Lastly, a knot may be tied and pushed down into the wound to close the trocar puncture site.

[0023] In an alternative procedure, the guide may be used to place a FIG. **8** stitch using two separate sutures rather than a single stitch using only one suture. The guide is initially inserted into the wound as previously described. A short tail at a first end of a first suture is loaded into the needle, passed through the first channel of the guide, and released into the abdominal cavity. At this point a short tail at the second end of the first suture is loaded into the needle, and passed through the second channel. The guide may then be rotated approximately 90 degrees from the initial orientation of the first suture passing. A short tail at one end of a second suture may be loaded into the needle, passed through the first channel of the guide, and released into the abdominal cavity. The needle is retracted and then a tail from the opposing end of the second suture is loaded and passed through the second channel of the guide and released into the abdominal cavity. The slider is pulled and the guide is removed from the wound with all four ends of suture captured in the snares. Knots may then be tied in each of the individual sutures to close the wound.

[0024] The basic procedural steps for abdominal entry and insufflation of the cavity may flow as follows. The needle is used to enter the abdominal cavity using standard closed laparoscopic techniques. A gas line is connected to the handle allow for an inert gas to be passed into the abdominal cavity. The inert gas is then turned on until the cavity reaches an appropriate level of insufflation to allow for the procedure to be performed with appropriate visualization. The needle is then removed, and a trocar is inserted into the puncture site to perform the procedure.

[0025] In an alternative embodiment, the guide may comprise a monolithic snare where the actuator section, shaft section and suture catchers, such as snare loops, are all integrally formed. The barrel may comprise two half barrel pieces that can be secured together. The barrel defines an inner lumen and at least two diagonal channels in communication with two corresponding window openings adjacent to a distal end of the barrel. Each channel comprises a wider proximal opening that tapers towards a distal exit so as to serve as a fulcrum for the needle apparatus directed therethrough. Each snare loop comprises a tip that may serve as a landmark indicator to position the guide against the internal peritoneal wall.

[0026] The snare loops can be completely retracted within the barrel by moving the actuator section and integral shaft section of the snare axially and proximally through the inner lumen. The enlarged tip of each snare loop comprises a positive stop abutting the narrower window opening within the barrel so as to prevent excess travel of the monolithic snare within the barrel.

[0027] The guide may comprise indicia on an outer barrel surface to indicate how far to insert the barrel into the abdomen (i.e., until the mark is no longer visible) so as to prevent the needle from entering the dermis.

[0028] An alternative method is provided to achieve a desirable placement of the needle and suture through the tissue, particularly for thinner abdominal walls. Prior to the placing the needle though the tissue, the lateral tip of a snare loop may be used as a landmark while tilting the guide at an angle to the primary guide axis, until the first lateral tip comes in close proximity to or abuts the peritoneum. The tissue area will compress, and placement of the needle will encompass a greater amount of tissue.

[0029] With the tilted guide in position, the suture engaging device, with a first free end section of suture engaged, may be inserted through a first channel while carrying the first section of suture. Once the needle exits the channel it passes through the compressed layers of tissue, and enters the body cavity. As it enters the body cavity, the needle passes through the first suture catcher, which may comprise a snare loop. The needle may release the strand of suture and be removed from the body leaving the suture section loosely inside the expanded snare loop. Thus, the suture section is carried into the body cavity to a point where the suture section intersects and traverses the generally planar opening defined by the expanded snare loop.

[0030] The guide is then tilted in the opposite direction, such that the opposite second lateral tip of the second suture catcher, such a second snare loop, comes in close proximity to the peritoneum. With the tilted guide in position, the second free end section of suture may then be engaged by the suture engaging device, and inserted through the opposing second channel to place the second end suture section. Once the needle exits the opposing channel it passes through the compressed layers of tissue and enters the body cavity. As it enters the body cavity, the needle passes through the opposing second suture catcher, e.g., snare loop. The needle may release the strand of suture and be removed from the body leaving the suture sections within the boundaries of the respective snare loops.

[0031] Moving the actuator section away from the barrel retracts the snare loops so as to capture the suture sections. With both suture sections captured, the guide is retracted from the tissue track, carrying the suture sections. With the guide and suture sections exposed outside the body cavity, the actuator section can be pushed toward the barrel extending the suture snare loops, and thus releasing the suture sections. A knot can then be tied in the suture sections and secured to provide closure of the wound.

[0032] Embodiments of the present invention provide a surgical guide device to be placed through a tissue track comprising a barrel defining a first channel and a second channel, the first channel comprising a first exit for directing a first suture portion into a body cavity, the second channel comprising a second exit for directing a second suture portion into the body cavity; a plurality of markings disposed along an exterior surface of the barrel, the plurality of markings operable to be disposed adjacent an inner abdominal wall to provide an active means to predictably pass a suture passer needle through tissue at a determined distance from an edge of a fascial defect; a distal tip located distally to the barrel and comprising a distal tip wall; a first self-supporting suture catcher positioned adjacent to the first exit; a second self-supporting suture catcher positioned adjacent to the second exit; and an actuator coupled to the first and second suture catchers, the actuator being configured to move the first and second suture catchers in unison between a radially extended configuration and a retracted configuration, wherein, in the radially extended configuration, the first self-supporting suture catcher defines a first unencumbered opening through which the first suture portion intersects the opening and then resides loosely within the first opening, wherein in the radially extended configuration, the second self-supporting suture catcher defines a second unencumbered opening through which the second suture portion intersects the opening and then resides loosely within the second opening; the first and second self-supporting suture catchers are apart from one another; and the first self-supporting suture catcher and the second self-supporting suture catcher extend radially outward and in opposite directions from the distal tip.

[0033] Embodiments of the present invention provide a surgical guide device to be placed through a tissue track comprising a barrel defining a first channel and a second channel, the first channel comprising a first exit for directing needle through tissue adjacent the barrel, the second channel comprising a second exit opposite the first exit for directing a needle through tissue adjacent the barrel, the channels operable for directing needles through tissue and a plurality of markings disposed along an exterior surface of the barrel, the plurality of markings operable to be disposed adjacent an inner abdominal wall to provide an active means to predictably pass a suture through tissue at a determined distance from an edge of a fascial defect; a distal tip located distally to the barrel and comprising a an aid to guide the suture to create a stitch across the defect.

[0034] Embodiments of the present invention provide a method of controlling a fascial bite of tissue when closing a trocar wound site comprising inserting a surgical guide device, as described herein, into the trocar site; visualizing the plurality of markings; aligning a selected one of the plurality of markings with the inner abdominal wall; inserting a needle through the first channel directing the needle through the tissue; inserting a needle through a second channel opposite the first channel, directing the needle through the tissue, a distal tip located distally to the barrel with an aid to guide the suture to create a stitch across the defect, wherein the fascial bite of tissue is set based on the selected one of the plurality of markings.

[0035] These and other features, aspects and advantages of the present invention will become better understood with reference to the following drawings, description and claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0036] Some embodiments of the present invention are illustrated as an example and are not limited by the figures of the accompanying drawings, in which like references may indicate similar elements.

[0037] FIG. **1** illustrates an oblique view of a needle according to an embodiment of the present invention;

[0038] FIG. **2** illustrates a perspective side view of a proximal end of the needle;

[0039] FIG. **3** illustrates a perspective side view of a distal end of the needle;

[0040] FIG. **4**A illustrates a perspective side view of a strand of suture being loaded into the needle with an extended capture rod;

[0041] FIG. **4**B illustrates a perspective side view of a strand of suture captured by the needle after retraction of the capture rod;

[0042] FIG. **5**A illustrates a cross-sectional view of the needle demonstrating the internal components;

[0043] FIG. **5**B illustrates a cross-sectional view of the needle demonstrating the inner workings of a needle handle;

[0044] FIG. **6**A illustrates an oblique view of the guide with radially expanded arms and suture catchers;

[0045] FIG. **6**B illustrates an oblique view of the guide with radially expanded arms and contracted suture catchers;

[0046] FIG. **6**C illustrates an oblique view of the guide with radially contracted arms and suture catchers;

[0047] FIG. **7** illustrates an oblique view of the distal tip of the guide with radially expanded arms and suture catchers;

[0048] FIG. **8** illustrates an oblique view of the guide reveals pathways for suture passing;

[0049] FIG. **9**A illustrates a cross-sectional view of the guide showing internal components;

[0050] FIG. **9**B illustrates an enlarged cross-sectional view of the guide showing distal internal components;

[0051] FIG. **9**C illustrates an enlarged cross-sectional view of the guide showing proximal internal components;

[0052] FIG. **10**A illustrates an exploded view of barrel tip components;

[0053] FIG. **10**B illustrates an isometric view of barrel tip components;

[0054] FIG. **11** illustrates a perspective side view of the insertion of guide into tissue;

[0055] FIG. **12** illustrates a perspective side view of the insertion of the needle with first end of suture strand into the guide;

[0056] FIG. **13**A illustrates a perspective side view of the insertion of the needle with second end of suture strand into the guide;

[0057] FIG. **13**B illustrates a perspective side view of an enlarged view of the insertion of the needle with second end of suture strand into the guide;

[0058] FIG. **13**C illustrates a perspective side view of the guide with the suture catchers retracted so as to capture the suture sections;

[0059] FIG. 13D illustrates a perspective side view of the guide retracted out from the tissue track with the captured suture sections;

[0060] FIG. 14 illustrates an isometric view of snare with protective basket;

[0061] FIG. 15 illustrates a preferred method for closing a surgical wound with a single stitch using a system comprising preferred embodiments of a guide and a suture engaging device disclosed above;

[0062] FIG. 16 illustrates a preferred method for closing a surgical wound with a figure eight stitch using the preferred system disclosed above;

[0063] FIG. 17 illustrates a method for creating pneumoperitoneum for laparoscopic procedures using an insufflation needle;

[0064] FIG. 18 illustrates an isometric view of the guide and suture engaging device;

[0065] FIG. 19A illustrates an exploded view of the guide;

[0066] FIG. 19B illustrates an isometric view of the guide;

[0067] FIG. 20A illustrates a detailed view of the proximal end of the guide;

[0068] FIG. 20B illustrates an isometric view of the guide with various positions of the suture engaging device;

[0069] FIG. 21A illustrates an isometric view of the guide with the snare retracted;

[0070] FIG. 21B illustrates a detailed view of the distal end of the guide with the snare retracted;

[0071] FIG. 22A illustrates a front view of the guide with snares extended;

[0072] FIG. 22B illustrates a planar view of the suture catchers;

[0073] FIG. 23 illustrates a detailed view of the proximal end of the guide;

[0074] FIG. 24 illustrates an isometric view of the guide and suture engaging device;

[0075] FIG. 25A illustrates front view of the guide and needle placed in the wound;

[0076] FIG. 25B illustrates detail view of the guide and needle placed in the wound;

[0077] FIG. 26 illustrates front view of the guide and needle placed in the wound with the needle piercing the skin;

[0078] FIG. 27A illustrates front view of the guide and needle placed deeper in the wound;

[0079] FIG. 27B illustrates detail view of the guide and needle placed deeper in the wound;

[0080] FIG. 28A illustrates isometric view of the guide placed deeper in the wound;

[0081] FIG. 28B illustrates detail view of the guide with the line mark positioned in the wound;

[0082] FIG. 28C illustrates detail view of the guide with the lateral tip positioned in the wound;

[0083] FIG. 29 illustrates isometric view of the guide and suture engaging device in the wound where the guide is tilted;

[0084] FIG. 30 illustrates an alternative method for closing a surgical wound with suture using a system comprising preferred embodiments of a suture engaging device and a guide having a monolithic snare;

[0085] FIG. 31 illustrates isometric view of the guide with a recess feature;

[0086] FIG. 32 illustrates a detailed view of the recess feature;

[0087] FIG. 33A illustrates a perspective view of a suture passer disposed within a guide, with the guide having graduated markings thereupon for depth and suture fascial tissue bite control;

[0088] FIG. 33B illustrates a detailed view of a suture passer moving through a suture catcher, with graduated markings applied to the barrel for depth control;

[0089] FIGS. 34A through 34C illustrate detailed views of using different depths, as controlled by visualization of a logo on the barrel, to control the fascial bite of tissue;

[0090] FIGS. 35A and 35B illustrate how the use of conventional equipment can result in a too large or too small distance for the suture; and

[0091] FIGS. 36A and 36B illustrate how the use of the suture passer system of the present invention results in consistent fascial bite of tissue.

[0092] The illustrations in the figures may not necessarily be drawn to scale.

[0093] The invention and its various embodiments can now be better understood by turning to the following detailed description wherein illustrated embodiments are described. It is to be expressly understood that the illustrated embodiments are set forth as examples and not by way of limitations on the invention as ultimately defined in the claims.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS AND BEST MODE OF INVENTION

[0094] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the term "and/or" includes any and all combinations of one or more of the associated listed items. As used herein, the singular forms "a," "an," and "the" are intended to include the plural forms as well as the singular forms, unless the context clearly indicates otherwise. It will be further understood that the terms "comprises" and/or "comprising," when used in this specification, specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

[0095] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one having ordinary skill in the art to which this invention belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0096] In describing the invention, it will be understood that a number of techniques and steps are disclosed. Each of these has individual benefit and each can also be used in conjunction with one or more, or in some cases all, of the other disclosed techniques. Accordingly, for the sake of clarity, this description will refrain from repeating every possible combination of the individual steps in an unnecessary fashion. Nevertheless, the specification and claims should be read with the understanding that such combinations are entirely within the scope of the invention and the claims.

[0097] In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It will be evident, however, to one skilled in the art that the present invention may be practiced without these specific details.

[0098] The present disclosure is to be considered as an exemplification of the invention and is not intended to limit the invention to the specific embodiments illustrated by the figures or description below.

[0099] As is well known to those skilled in the art, many careful considerations and compromises typically must be made when designing for the optimal configuration of a commercial implementation of any system, and in particular, the embodiments of the present invention. A commercial implementation in accordance with the spirit and teachings of the present invention may be configured according to the needs of the particular application, whereby any aspect(s), feature(s), function(s), result(s), component(s), approach (es), or step(s) of the teachings related to any described embodiment of the present invention may be suitably omitted, included, adapted, mixed and matched, or improved and/or optimized by those skilled in the art, using their average skills and known techniques, to achieve the desired implementation that addresses the needs of the particular application.

[0100] Broadly, embodiments of the present invention provide a system for closing trocar wound sites that includes a suture engagement device and a guide to direct the device through the body tissues. An improved needle and guide instrument set, as further described below, allows for the surgical placement of suture to be completed "blind," namely, without the aid of an endoscope for direct visualization of the abdominal cavity. This may be advantageous in certain surgical procedures where an endoscope is not being used or does not provide adequate visualization of the surgical site.

[0101] A preferred embodiment of a suture engagement device or suture passing needle, or simply needle, **100** is shown in FIG. **1**. The needle **100** may simply have a handle **101** and actuator **105** at the proximal end **12**. An outer tube **108** is connected to the handle **101** that terminates with a suture capture mechanism **107**, sharp needle tip **109**, and spring loaded safety tip **112** at the distal end **14** of the needle **100**, as shown in FIG. **3**.

[0102] FIG. **2** illustrates a closer view of the proximal portion **16** of the needle **100**. The base of the proximal end **12** of the needle **100** is a main handle or housing **101**. The handle **101** may have a finger loop **102** that may accommodate one or more fingers, as well as a series of one or more grooves **103** to accommodate the placement of additional fingers. The finger loop **102** and grooves **103** may provide the user with a comfortable, secure grip of the device and provide greater control when handling the device **100**. Slidably disposed within the housing **101** is an actuator **105** that may be used to control a suture capture mechanism **107** at the distal tip of the needle **101**. At the proximal end of the actuator **105**, an enlarged surface **106** may provide an ergonomic location for a thumb or other finger to trigger the actuator **105**. The length travel of the actuator **105** may be constrained by a pin **123** that connects to the actuator **105** and slides within a slot **104** on the handle **101**. The pin **123** also serves the purpose of rotationally constraining the actuator **105**.

[0103] FIG. **3** illustrates a closer view of the distal portion **18** of the needle **100**. An obturator tube **111** may be slidably disposed within an outer tube **108**, and a capture rod **114** may be slidably disposed within the obturator tube **111**. The outer tube **108** terminates at a sharp tip **109** that may have two or more beveled edges **118** to facilitate the ease of passage of the needle **100** through tissue. The obturator tube **111** has a blunt surface **112** at its distal end **20** and may be used to serve as a safety tip for the needle **100**. The obturator tube **111** may be spring loaded such that it can passively travel between an axially extended and retracted position. Initially, the obturator tube **111** may be biased in the extended position where the blunt surface **112** extends further distally than the sharp tip **109** of the outer tube **108**. As the distal tip of the needle **100** is pushed into the tissue with enough load to overcome the force of the spring, the obturator tube **111** may retract proximally, ultimately exposing the sharp tip **109** of the outer tube **108**. The sharp tip **109** and edges **118** may then minimize the trauma to the tissue layers as the needle **100** is inserted. Once the tip **109** of the needle **100** enters the body cavity, the obturator tube **111** may passively return to the distally extended position shielding the sharp tip **109** from inadvertently damaging the tissue structures within the body cavity. The purpose of the capture rod **114** is to secure the suture to the needle **100**, in a manner further described below.

[0104] In FIGS. **4**A and **4**B, the suture capture mechanism **107** at the distal end **14** of the needle **100** preferably comprises a channel **110** in the outer tube **108**, a cutout section **113** of the obturator tube **111**, and a slot **115** in a capture rod **114**. The suture capture mechanism **107** may function through the movement of the capture rod **114** between an axially extended and retracted position that is controlled by the actuator **105** at the proximal end **12** of the needle **100**. In FIG. **4**A, the capture rod **114** is shown in the extended position such that a strand of suture **125** can be placed through the cutout **113** in the obturator **111** and into the slot **115** of the capture rod **114**. In this axially extended position, the obturator cutout **113** is open to, and aligned with, the capture rod slot **115** in order to receive the suture **125**. In FIG. **4**B, the capture rod **114** is shown in the axially retracted position such that the strand of suture **125** is secured between the channel **110** of the outer tube **108** and the distal face **117** of the slot **155** in the capture rod **114**. When the suture **125** needs to be released, the actuator **105** returns the capture rod **114** to the extended position. A proximal face **116** of the capture rod **114**, shown in FIG. **4**A, may aid in pushing the suture **125** out of the obturator cutout **113** and away from the outer profile of the outer tube **108**.

[0105] Cross sectional views of the preferred embodiment of the needle **100** are shown in FIGS. **5**A and **5**B that illustrate the inner workings of the handle **101**. A proximal portion **22** of the obturator tube **111** may be connected to an obturator hub **119**. The hub **119** may have one or more rotational alignment structures such as a flat face **120** or an alignment post **121** that prevent the hub **119** from rotating about a long axis **24** as it translates back and forth. Additional extrusions **128** inside the housing **101** may be necessary to mate with the alignment protrusions **120**, **121** on the hub **119**. The alignment of the hub **119** is critical to ensure the cutout **113** at the distal end maintains its alignment for suture capturing. The hub **119** may also have a cylindrical portion **126** to accommodate a coil spring **122** that is used to spring load the obturator tube **111**. The opposing end of the spring **122** may connect to the actuator **105** at a similarly accommodating cylindrical portion **127**. The capture rod **114** may be attached to the actuator **105** with a pin **123** that passes through a hole **124** in the actuator and a slot **129** in the capture rod. As previously described, the pin **123** may slide within a window **104** of the housing **101** that is used to limit the travel and prevent rotation of the capture rod **114**.

[0106] In FIGS. **6A-6C**, a preferred embodiment of a guide **130** may be useful for directing the above described suture engaging device through a body wall. The guide **130** may be particularly useful for the placement of sutures used in closing wounds or openings through body walls made in surgical procedures to access internal body cavities. Accordingly, the guide **130** preferably comprises two pathways diagonal to each other and oriented to direct a needle apparatus to both a first internal location to carry and release a first end of a suture, and a second internal location to carry and release a second end of a suture.

[0107] An oblique view of the preferred guide **130** is shown in FIGS. **6A-6C**. The guide **130** may comprise a distal barrel tip **140** with two radially expanding arms **147**, two suture capturing mechanisms, or suture catchers, which may comprise snare loops **146** in this illustrated embodiment, a main barrel **131** with two channels **132**, **136** (shown in FIG. **8**), a plunger **153**, and a slider **156**. The two channels **132**, **136** are disposed within the main barrel **131** to guide the suture passing needle **100** through the tissue to be sutured, as shown in FIGS. **8** and **12**. The slider **156** may be slidably disposed onto the main barrel **131** to provide actuation of the suture snare loops **146**. The plunger **153** may be slidably disposed onto the main barrel **131** to provide actuation of the expanding arms **147**. The expanding arms **147** may serve as an internal cavity securing mechanism, which is used to secure the guide **130** against the internal peritoneal wall. The two suture snare loops **146** serve the purpose of capturing the suture material after it has been passed through the tissue wall. The slider **156** may facilitate the ease of handling of the guide **130**.

[0108] As the slider **156** translates with respect to the barrel **131**, the suture snare loops **146** move between two positions of radially extended shown in FIG. **6**A and retracted shown in FIG. **6**B. When each snare **146** is in the radially extended configuration, a loop, ring, circle or hoop shape is preferably formed. The snare loop **146** provides a region for a suture passing needle to pass into and drop off the suture. The snare loop **146** defines an opening through which the carried suture traverses. When the carried suture is released from the suture engaging device, the suture section intersecting the opening of the snare loop **146** resides loosely until the snare loop **146** is retracted. When the snare loop **146** is retracted the suture material becomes trapped between the snare loop **146** and the wall of the distal barrel tip **140**.

[0109] Further, as the plunger **153** translates with respect to the barrel **131**, the radially expanding arms **147** at the distal barrel tip **140** move between two positions of radially expanded, or flared out, as shown in FIG. **6**B, and radially contracted, or slender, as shown in FIG. **6**C. Each of the expanding arms **147** preferably comprise a living hinge section, where the material is cut thin at specific locations allowing for the material to flex. It is to be expressly understood that the expanding arms may comprise a variety of structures and mechanisms that are capable of moving between slender and flared out configurations.

[0110] As shown in FIG. **7**, radially adjacent to the expanding arms **147** may be one or more distally extending stop tabs **143**. The stop tabs **143** are configured to provide a mechanical stop for the distal end cap **141** of the barrel tip **140** to collide against preventing excessive flex within the thin sections **148**, **149**, **150** of the expanding arms **147**. The stop tabs **143** may also have a slot **145** cutout in the outer wall that provide a region for the snare loop material **146** to be housed when the snare loop **146** is in the retracted state. When a section of suture **159** is captured within the snare loop **146**, the slot **145** may provide additional security to the captured suture **159**, as opposed to the snare loop **146** pulling the suture against a smooth surface.

[0111] The barrel tip **140** may comprise a separate component that is assembled to the main barrel **131**. Two tabs **166** on the main barrel **131** may be placed radially opposite to each other. Slots **167** in the barrel tip **140** may allow for the tip **140** to initially slide past the tabs **166** on the main barrel **131**. The slots **167** may have an undercut section **168** that may be engaged by rotating the tip **140** and pulling it distally. The gap that is created between the tip **140** and the barrel **131** may be filled with a deformable c-shaped clip **169** to prevent the tip **140** from dislodging from the barrel **131**.

[0112] The slider **156** is slidably disposed on the main barrel **131** as it can travel between a distal position shown in FIG. **6**A and a proximal position shown in FIG. **6**B. Two finger tabs **154** are preferably disposed near the distal end of the slider **156** and configured to be grasped with one or more fingers. The thumb or palm of the hand may then be pressed against the proximal face **157** of the plunger **153**. The finger tabs **154** may then be pulled proximally to actuate the guide **130** with the plunger **153** being used to provide a counterforce. The slider **156** can reach a stop, achieved by either an internal feature of the slider **156** engaging the main barrel **131** or the snare loops **146** being fully retracted. Additional force between the slider **156** and the plunger **153** may overcome the counterforce translated to the plunger **153** by a spring. The plunger **153** can then move axially to actuate the expanding arms **147** to the slender contracted state. Upon release of the force between the slider **156** and the plunger **153**, the plunger **153** can return to the original position and result in the expanding arms **147** reverting to the radially expanded position. The slider **156** may be biased to the distal position by a spring that passively returns the slider **156** to the distal position when released, thus resulting in the suture loops **146** reverting back to the radially extended position. It may be appreciated that biasing the guide **130** in this configuration may allow for the surgeon to have his hands free from operating the guide **130** such that work with other instruments may be conducted once the guide **130** has been inserted into the trocar wound.

[0113] In the preferred embodiment, the guide **130** provides two different, diagonal pathways for a suture passing needle apparatus and thus comprises first and second channels **132**, **136** as more clearly shown in FIG. **8**. The channels **132**, **136** are directionally diagonal to the long axis "**24**" of the guide **130** at some acute angle, preferably in the range of 5-30 degrees. The angle of the channels **132**, **136** controls the depth of bite into the tissue, such that a greater angle provides a greater bite of tissue. Angular spacing between entries to the channels **132**, **136** are preferably equal (i.e., equiangular) depending upon the number of channels, e.g., 180 degrees apart if there are two channels, 120 degrees apart if there are three channels, etc. Two separate entry points **133**, **137** are provided on the channels **132**, **136** for a suture passing needle to enter the guide **130**, as well as two separate exit points **134**, **138**. Each channel **132**, **136** has a slot **135**, **139**, or opening that allows for a strand of suture to be removed from the channel. This may be necessary to complete the suture loop in the tissue.

[0114] The internal workings of the guide **130** are shown in FIGS. **9**A-9C, which illustrate cross-sectional views through the center of the guide **130**. An inner shaft **158** spans from the proximal end **30** to the distal end **32** of the guide **130** and is used to actuate the expanding arms **147**. The shaft **158** connects at the distal end **32** of the guide **130** to the distal end cap **141** of the barrel tip **140**, and connects at the proximal end **30** of the guide **130** to the plunger **153**. A retaining ring, heat stake, or other mechanical fastening means may be used to secure the shaft **158** to the plunger **153**. When the slider **156** is pulled into the proximal position as shown in FIG. **6**B, a counterforce is applied to the plunger **153**. Upon the slider **156** reaching the full proximal extension, further pushing on the plunger **153** advances the shaft **158** distally which in turn advances the end cap **141** on the barrel tip **140** distally, causing the expanding arms **147** to radially contract as shown in FIG. **6**C. Upon release of the plunger **153** and slider **156**, the shaft **158** may passively return to its native proximal position, pulling the end cap **141** with it and causing the expanding arms **147** to radially extend.

[0115] Two springs are used to bias the guide **130** into the configurations shown in FIGS. **6**A and **9**A, with the radially extended arms **147** and snare loop **146** expanded, and the plunger **153** in the proximal position. A slider spring **155** is used to bias the slider **156** to the proximal position. In FIG. **9**C, one end of the slider spring **155** is captured at the proximal end **40** of the main barrel **131** by two spring capture tabs **173**. The opposing end of the spring **155** rests against an inner edge **174** of the slider **156**, forcing the slider **156** to a proximal position. In FIG. **9**B, a plunger spring **165** is used to bias the distal barrel tip **140** in the proximal position. The spring **165** is captured between inner shaft **158** and the proximal face of track support **170** connected to the barrel tip **140**. This plunger spring **165** forces the inner shaft **158** and plunger **153** in a proximal position, pulling the barrel tip **140** proximally as well.

[0116] In FIG. **9**C, the free ends of a snare cord loop **146** may be captured in a retaining cap **175** that is fixed to the proximal end of the slider **156**. The snare cord **146** may traverse distally down the guide **130** either through snare channels **176** inside the main barrel **131** or guide tubes **171** that span through a portion of the length of the main barrel **131**. The snare channels **176** or guide tubes **171** may be used to constrain the snare cord material **146** and prevent it from bunching up or buckling under load as it travels proximally and distally within the guide **130**.

[0117] FIGS. **10**A-10B illustrate the laminate component construct that provides tracks **172** within the barrel tip **140**. The track support **170** is positioned between two track side panels **177**. The tracks **172** may be used to guide the snare cord **146** from a position within the barrel tip **140**, parallel to the long axis of the guide **130** to a position exiting the barrel tip **140** relatively perpendicular to a long axis **42** of the guide **130**. These curved exit tracks **172** may take on an angulation of approximately 90 degrees, however this angle may be made more acute or obtuse to optimize the positioning of the snare.

[0118] In the case of a laparoscopic surgery involving use of a trocar, the guide **130** may be placed through the tissue layers of the open trocar wound site as shown in FIG. **11**. This tissue track may consist of skin **160**, adipose tissue **161**, muscle and fascia **162** and the peritoneum **163**. Prior to insertion of the guide **130** through the tissue track, the slider **156** and plunger **153** may be pulled into the compressed position to place the arms **147** in the slender configuration, and the snare loops **146** in the retracted position. Once the distal barrel tip **140** of the guide **130** is appropriately placed posterior to the peritoneal layer **163**, the slider **156** and plunger **153** may be released and spring-biased to the open position, extending the suture snare loops **146** and expanding the tissue engaging arms **147**. The guide **130** may then be retracted until the tissue engaging arms **147** are resting against the inner peritoneal wall **163** to align the channels **132**, **136** with the appropriate layers of tissue to be sutured.

[0119] Once the guide **130** is secured against the peritoneal wall **163**, the suture engaging device **100**, with a first free end section **44** of suture **125** engaged, may be inserted through a channel **132** while carrying the section **44** of suture **125**, as shown in FIG. **12**. Once the needle **100** exits the channel **132** it passes through various layers of tissue **162**, **163** and enters the body cavity. As it enters the body cavity, the needle **100** passes through the snare loop **146**. The needle **100** may release the strand of suture **125** and be removed from the body leaving the suture section **44** loosely inside the expanded snare loop **146**. Thus, the suture section **44** is carried into the body cavity to a point where the suture section **44** intersects and traverses the generally planar opening **48** (see FIG. **11**) defined by the expanded snare loop **46**.

[0120] The second free end section **46** of suture **125** may then be engaged by the suture engaging device **100**, and inserted through the opposing channel **136** to place the second end suture section **46**. As shown in FIGS. **13**A-13B, once the needle **100** exits the channel **136** it passes through various layers of tissue **162**, **163** and enters the body cavity. As it enters the body cavity, the needle **100** passes through the opposing snare loop **146**. The needle **100** may release the strand of suture **125** and be removed from the body leaving the suture sections **44**, **46** within the boundaries of the respective snare loops **146**.

[0121] FIG. **13**C shows the snare loops **146** retracted so as to capture the suture sections **44**, **46**, which is the configuration actuated in FIG. **6**B when the slider **156** is moved proximally with respect to the main barrel **131**. With both suture sections **44**, **46** captured, the guide **130** is ready to be retracted from the tissue track **50**, carrying the suture sections **44**, **46** as shown in FIG. **13**D.

[0122] FIG. **14** illustrates an alternative embodiment including a pouch, or basket, **178** secured to the snare. The pouch **178** is positioned below the snare and comprises a floor **180** as a protective element to prevent the distal tip of the needle **100** from extending deeper into the abdominal cavity. The pouch may be constructed of a compliant fabric, rigid polymer, or other such material to obstruct needle penetration.

[0123] In FIG. **15** a preferred method **200** is disclosed for closing a surgical wound using a surgical instrumentation system comprising a guide and suture engagement device as described above. It will be appreciated that this method **200** enables an operator to pass and retrieve the suture using simply a guide and suture engaging device without the need for additional instrumentation or visualization inside the body cavity. The initial step **201** comprises actuation of the guide into a slender configuration and insertion of the distal tip into the surgical wound. In step **202**, a securing mechanism, which may comprise living hinges, is radially expanded for engagement with the inner body cavity wall when the guide is retracted, preventing the guide from being pulled out of the wound as well as providing a reference point for the suture placement. Step **203** comprises the deployment of two suture catchers, which may preferably comprise snare loops, to be used to capture the suture material. It can be appreciated that step **202** and **203** may be combined such that they occur simultaneously. It will further be appreciated that steps **202** and **203** may occur by default (e.g., through use of springs) upon release of the guide such that the operator may have both hands free to engage other instruments.

[0124] Step **204** comprises engaging and capturing a first section of a suture with a suture capture mechanism disposed at the distal end of a suture engaging device having a shaft. In step **205** the suture engaging device with secured suture is inserted through a first track of the guide, through various tissue layers, and ultimately ending inside the body cavity as it passes through the first snare loop. Step **206** comprises releasing the first suture section from the suture engaging device and retracting the device from the body and guide. The first section of the suture may remain inside the body cavity and loosely encapsulated within the boundaries of the expanded first snare loop.

[0125] Step **207** comprises engaging and capturing a second section of the same suture with the suture capture feature on the suture engaging device. In step **208**, the suture engaging device with secured suture is inserted through a second track of the guide, through various tissue layers, ending inside the body cavity as it passes through the second snare loop. Step **209** comprises releasing the suture from the suture engaging device and retracting the device from the body and guide while leaving the second section of the suture inside the body cavity and loosely encapsulated within the boundaries of the second snare loop.

[0126] In step **210**, the suture material may be passed through the slots in the channels on the guide to release the suture from the constraint of the channel. Step **210** comprises retracting the snare loops back against the guide and securing the suture between the outer guide wall and the snare loop. In step **211**, the guide may be actuated such that the securing mechanism is radially contracted and converting the guide into the slender configuration. It can be appreciated that step **210** and **211** may be combined such that they occur simultaneously or seamlessly with a single motion. In step **213** the guide can be removed from the body cavity with the two captured sections of the suture. Step **214** comprises releasing the suture from the snare loops on the guide. The procedure may then be completed at step **215** by forming a single stitch loop to close the surgical wound.

[0127] In FIG. **16** an alternative method **300** is illustrated for closing a surgical wound using a surgical instrumentation system by forming a figure eight stitch as opposed to the single stitch method **200**. In step **301** a first length of suture is passed into the body at a first location of the wound comprising the same steps **201-210** as previously described. In step **302**, the guide is rotated 90 degrees from the initial position in step **301**. In step **303**, a second length of suture is passed into the wound at a second location of the wound comprising steps **202-214**. Step **304** comprises completing the closing of the wound by forming a stitch loop with each of the suture strands.

[0128] In FIG. **17** a preferred method **400** is illustrated for creating pneumoperitoneum for laparoscopic procedures using an insufflation needle. Step **401** comprises inserting the tip of the needle through the abdominal wall and into a body cavity. This step may also initially require a small incision in the skin to be made prior to insertion of the needle. In step **402**, a gas line may be connected to the needle. Step **403** comprises pumping an inert gas through the needle and into the body cavity until an appropriate internal pressure is achieved. Lastly, in step **404** the needle may be removed from the body and the endoscopic procedure can begin.

[0129] In FIG. **18**, an alternative embodiment of a guide **502** may be useful for directing a suture engaging device **501** through a body wall. The guide **502** may be particularly useful for the placement of sutures used in closing wounds or openings through body walls made in surgical procedures to access internal body cavities. Accordingly, the guide **502** preferably comprises two pathways diagonal to each other and oriented to direct a needle apparatus to both a first internal location to carry and release a first end of a suture, and a second internal location to carry and release a second end of a suture. The guide **502** comprises a proximal end **581** and a distal end **582**.

[0130] An oblique view of the preferred guide **502** is shown in FIGS. **19**A-**19**B. The guide **502** may comprise a monolithic snare, or suture capturing mechanism, **504**. The suture capturing mechanism may comprise two suture catchers **507**. In a preferred embodiment, each suture catcher **507** may comprise a self-supporting snare loop **507**. The guide **502** further comprises a shaft section **506**, and an actuator section **505**, all integrally formed. A barrel **503** comprises two channels **509**, **513** as shown in FIG. **20***a*, and a clip **510**. The barrel **503** also defines an inner lumen **520** and two or more window openings **512**, **521**. The barrel **503** preferably comprises two half barrel pieces **514**, **515** secured to one another. The half barrel pieces **514**, **515** may be identical in geometry. The barrel **503**, which may be formed by securing the two half barrel pieces **514**, **515** together, defines an inner lumen **520** that extends along an axis A of the guide **502**, and two or more channels **509**, **513** diagonal to the axis A. Each channel **509**, **513** is in communication with a corresponding window opening **512**, **521** located adjacent to the distal end **582**. Ultrasonic welding, adhesive bonding, snap leg features, or a clip **510**, or a number of other means may be used to secure the half barrel pieces **514**, **515** together. The two channels **509**, **513** preferably formed by the construct of the barrel half pieces **514**, **515**, are disposed within the barrel **503** to guide the suture passing needle **501** through the tissue to be sutured. The shaft section **506** of the monolithic snare **504** may be slidably disposed within the inner lumen **520** of the barrel **503** to provide actuation of the suture snare loops **507**. The snare loops **507** may traverse within and extend laterally from the window openings **512**, **521**.

[0131] The tips **516** of the snare loops **507** may serve as a landmark indicator, which is used to position the guide **502** against the internal peritoneal wall. The two suture snare loops **507** serve the purpose of capturing the suture material after it has been passed through the tissue wall. The actuator section **505** of the monolithic snare **504** and tabs **508** of the barrel **503** may facilitate the ease of handling of the guide

502. Each snare loop **507** preferably comprises a pre-formed shape with memory characteristics such that the snare loop **507** consistently takes on the same shape whenever fully expanded.

[0132] As the actuator section **505** and shaft section **506** translate with respect to the barrel **503**, the suture snare loops **507** move between two positions of radially extended shown in FIGS. **19**A and **19**B, and retracted shown in FIG. **21**A. When each snare loop **507** is in the radially extended configuration, a pre-defined loop, ring, circle or hoop shape preferably extends laterally from the barrel **503**. In the preferred embodiment, each snare loop **507** provides a corresponding C-shaped region **511** (from a side view as shown in FIG. **22**A) for a suture passing needle to pass into and drop off the suture. Each region **511** is concave with respect to the proximal end **581** of the guide **502**. In the preferred embodiment, each snare loop **507** is aligned with a corresponding tab **508** located directly above. Each snare loop **507** defines an opening through which the carried suture traverses. In the preferred embodiment having a pair of snare loops **507**, each snare loop **507** is preferably spaced 180 degrees apart from the other and disposed distally adjacent to a corresponding channel exit point **518**.

[0133] When the carried suture is released from the suture engaging device, the suture section intersecting the opening of the snare loop **507** resides loosely until the snare loop **507** is retracted. When the snare loop **507** is retracted the suture material becomes trapped between the snare loop **507** and the wall of the barrel **503** at the window opening **512** in the barrel (see, e.g., FIG. **13**C). In this retracted position, the actuator section **505** is pulled away from the barrel as shown in FIG. **21**A. The projecting snares loops **507** are collapsed as they are withdrawn into the window opening **512** and further into the inner lumen **520** of the barrel **503**. The projecting sections of each snare loop **507** converge at the distal end to form a rigid tip **516** that points upwardly and outwardly from the barrel **503**. The window opening **512** provides a recess for housing the tip **513** to allow the tip **516** from protruding outside the barrel **503** outer profile when in the retracted position, thus enabling smooth entry of the guide when being placed through the tissue track. As shown in FIG. **21***b*, the window opening **512** decreases in size as it extends towards the inner lumen **520**. The snare tip **516** has an enlarged size or width that acts as a positive stop in combination with the narrowing window opening **512** to prevent excess proximal travel (i.e., retraction) of the monolithic snare **504** within the barrel **503**.

[0134] The monolithic snare **504** may be constructed from a polymer that can withstand a high degree of strain. The percent of elongation at yield may range from 3-8%. The snare loop **507** may have thin cross-sections to provide the flexibility to easily bend and conform to various geometric shapes yet stiff enough to create a self-supported snare loop that extends generally perpendicular to the long axis of the guide. The monolithic snare **504** may also extend laterally in a geometry that from a side view is a continuous radius **525** as shown in FIG. **22**A. The continuous radius **525** effectively keeps the rigid tip along the "continuous radius" path as the snare loop **507** is being retracted into the barrel. The ratio of the continuous radius **525** to the cross sectional geometry of the snare loop **507** is such that the strain level of the material when retracted and constrained in the barrel **503** will not cause noticeable permanent deformation of the memorized pre-formed shape when constrained for a duration up to one hour. The same principle is true for the snare radii **526**, **527**, **528** as shown in the top planar view illustrated in FIG. **22**B. Materials that may be used to construct the monolithic snare **504** include plastics such as nylon, polyethylene, polyester or polypropylene.

[0135] In this alternative embodiment, the guide **502** provides two different, diagonal pathways for a suture passing needle apparatus and thus comprises first and second channels **509**, **513** as more clearly shown in FIG. **20**A. The channels **509**, **513** are directionally diagonal to the long axis A of the guide **502** at some acute angle, preferably in the range of 5-30 degrees. The angle of the channels **509**, **513** controls the depth of bite into the tissue, such that a greater angle provides a greater bite of tissue. Angular spacing between entries to the channels **509**, **513** are preferably equal (i.e., equiangular) depending upon the number of channels, e.g., 180 degrees apart if there are two channels, 120 degrees apart if there are three channels, etc. Therefore, channel exits are also preferably equiangular. Two separate entry points **521**, **522** are provided on the channels **509**, **513** for a suture passing needle to enter the guide **502**, as well as two separate exit points **518**, **523**. Each channel **509**, **513**, has a corresponding slot **512**, **524**, or opening that allows for a strand of suture to be removed from the channel. This may be necessary to complete the suture loop in the tissue. As shown in FIGS. **20**A-B, the proximal section of each channel **509**, **513**, may have a wider mouth including a tapered section **517** that enables angulation of the suture passing needle **501**, such that a lesser angle allows a lesser bite of tissue.

[0136] The actuation section **505** of the monolithic snare **504** may have a locking leg **519** that snap fits into a latch **529** in the barrel **503** as shown in FIG. **23**. The locking leg **519** can be depressed to flex from the "locked" position to enable movement of the monolithic snare **504**. The actuation section **505** may also include a variety of other connectors to facilitate a releasable locking connection between the monolithic snare **504** and the barrel **503**.

[0137] In another embodiment shown in FIG. **24**, the guide **530** comprises a barrel **531** with indicia and a monolithic snare **532** with a ring-shaped actuation section **533** for easy manipulation of the monolithic snare **532**. The guide **530** includes two indicators to facilitate deployment; 1) a line mark **535** on an outer barrel surface **583**, and 2) a lateral tip **536** of each snare loops **534**. The two indicators can be employed in a method to accommodate proper positioning of the guide **530** in various abdominal wall thicknesses to achieve predictable suture placement in the tissue.

[0138] In the case of a laparoscopic surgery involving use of a trocar, the guide **530** may be placed through the tissue layers of the open trocar wound site as shown in FIGS. **25**A and **25**B to properly position the guide **530** within the body cavity for receiving suture carried by one or more suture engaging devices. This tissue track in the abdominal wall may include skin **537**, adipose tissue **538**, muscle and fascia **539** and the peritoneum **540**. Prior to insertion of the guide **530** through the tissue track, the actuator section **533** may be pulled away from the barrel **530** in a proximal direction, moving the snare loops **534** into the fully retracted position to achieve the most slender profile. Once the distal end of the guide **530** is appropriately placed posterior to the peritoneal layer **540**, the actuator section **533** may be pushed toward the barrel **531**, extending the suture snare loops **534**. The guide **530** may then be retracted, or translated in a proximal direction away from the body cavity to position the extended snare loops **534** for receiving suture carried by the suture engaging device and to align the channels with the appropriate layers of tissue to be sutured, as shown in FIG. **25***a*. For patients with a relatively greater abdominal wall thickness, proximally translating the guide **530** until the expanded snare loops abut the inner peritoneal wall **540** will concurrently position both channel exits beneath the skin surface **537**.

[0139] Once the guide **530** is positioned with the peritoneal wall **540**, the suture engaging device **501**, with a first free end section of suture **541** engaged, may be inserted through a first channel **544** while carrying the first section of suture **541**, as shown in FIGS. **25**A and **25**B. Once the needle **551** exits the channel **544** it passes through various layers of tissue **539**, **540**, and enters the body cavity **570**. As it enters the body cavity **570**, the needle **551** passes through the first snare loop **534**. The needle **551** may release the strand of suture **541** and be removed from the body leaving the suture section loosely inside the expanded first snare loop **534**. Thus, the suture section **542** is carried into the body cavity to a point where the suture section intersects and traverses the generally planar opening defined by the expanded first snare loop **534**.

[0140] The second free end section of suture **541** may then be engaged by the same suture engaging device **501** or a second engaging device, and inserted through the opposing second channel to place the second end suture section **543**. Once the needle **551** exits the opposing second channel it passes through various layers of tissue **539**, **540** and enters the body cavity **570**. As it enters the body cavity **570**, the needle **551** passes through the opposing second snare loop **534**. The needle **551** may release the second strand of suture **541** and be removed from the body leaving the suture sections **542**, **543** within the boundaries of the respective snare loops **534**.

[0141] Moving the actuator section **533** away from the barrel retracts the snare loops **534** so as to capture the suture sections **542**, **543**. With both suture sections **542**, **543** captured, the guide **530** is retracted from the tissue track, carrying the suture sections **542**, **543** and directing the suture in the inner abdominal space to create a stitch. With the guide **530** and suture sections **542**, **543** exposed outside the body cavity, the actuator section **533** can be pushed toward the barrel **531** extending the suture snare loops **534**, and thus releasing the suture sections **542**, **543**. A knot can then be tied in the suture sections **542**, **543** and secured to provide closure of the wound.

[0142] FIG. **25**B shows the placement of the needle **501** piercing through the tissue. The distance **545** from the puncture location of the needle **501** through the peritoneum **540** to the closest edge, or perimeter, of the defect, which coincides with the sidewall of the barrel, is desirable to be nominally 8 mm. With the guide **530** positioned perpendicular to the skin surface, the channels **544** are aligned to guide the needle **551** through the desirable amount of tissue to be sutured. However, for this method, the abdominal wall thickness **546** must be at an adequate thickness to assure the needle **551** will exit the channel **544** below the outer skin surface, or simply skin, **537** surrounding the body cavity. It is undesirable to have the suture **541** through the skin **537**.

[0143] FIG. **26** shows an abdominal wall thickness **547**, where for the above described method of positioning the lateral tips **536** resting against the peritoneal wall **540** with the guide perpendicular to the skin surface, the thickness is not adequate. The needle is shown with a puncture **548** through the skin **537** because the channel exits are positioned above or outside the skin **537**. The guide **530** needs to be inserted further into the wound for the needle **551** to avoid piercing the skin **537**—namely, for the needle **551** to exit the channel exits of the guide **530** subcutaneously. An indicator in the form of a line mark **535** is provided on the outer surface of the barrel **531** as a reference. When the line mark **535** is positioned below the skin surface **537**, the exit of each of the first channel **544** (shown) and second channel (not shown) will be below the outer skin surface **537** such that the needle **551** will not pierce through the skin **537**. FIG. **27**A shows the guide **530** inserted further in the wound, where the line mark **535** is below the skin surface **537** and thus no longer visible, such that both channel exits are positioned beneath the outer skin surface **537**. This results in the lateral tips **536** being a significant distance **548** away from the peritoneum **540**, thereby improperly positioning the guide **530** within the cavity. FIG. **27**B shows the placement of the needle **501** piercing through the tissue. The relatively shorter distance **545** from the puncture location of the needle **501** through the peritoneum **549** to the edge of the defect, which coincides with the sidewall of the barrel, may not be desirable in such thinner abdominal walls.

[0144] An alternative indicator could be a recess or protrusion along the outer guide surface to provide a tactile feedback when the guide is translated to a position where the recess or protrusion engages the skin. FIG. **31** shows a preferred embodiment of the guide **530** with a proper depth indicator **584** comprising a recess **584**. The recess **584** may be formed partially about the circumference of the barrel surface **583**. Furthermore, the edge of the recess may have a radius **585** at one or both ends of the edge, as shown in FIG. **32**. The radius **585** would facilitate the penetration of the barrel **531** past the skin surface with a twisting motion while applying translational force. The tactile feedback of the recess **584** abutting the skin informs the user the guide is positioned such that both channel exits are positioned beneath the outer skin surface **537**. An indicator feature, such as the extended lateral tips **536** of the snare loops, may then be utilized to determine the guide position relative to the peritoneum **540**, An alternative feature could be a mark on the barrel body at a level equivalent to the lateral tips of the snare. If greater guide depth is desired, the guide may be twisted while applying a translation force to penetrate the guide beyond the skin surface.

[0145] An alternative method can be used to achieve a desirable placement of the needle **551** and suture **541** through the tissue. FIG. **28**A shows the guide **530** positioned in a relatively thinner abdominal wall thickness **547** as described above. FIG. **28**B shows the line mark **535** below the skin **537**. FIG. **28***c* shows the lateral tip **536** a vertical distance **548** from the peritoneum **540**. Prior to the placing the needle **551** though the tissue **539**, **540**, the lateral tip **536** may be used as a landmark while tilting the guide **530** at an angle to the primary guide axis **560**, until the first lateral tip **536** comes in close proximity to or abuts the peritoneum **540**, as shown in FIG. **29**. The tissue area **562** will compress, and placement of the needle **551** will encompassed a greater amount of tissue **539**, **540**.

[0146] With the tilted guide **530** in position, the suture engaging device **501**, with a first free end section **542** of suture **542** engaged, may be inserted through a first channel

544 while carrying the section **542** of suture **541**. Once the needle **551** exits the channel **544** it passes through the compressed layers of tissue **539**, **540**, and enters the body cavity. As it enters the body cavity, the needle **551** passes through the first snare loop **534**. The needle **551** may release the strand of suture **541** and be removed from the body leaving the suture section **542** loosely inside the expanded snare loop **534**. Thus, the first suture section **542** is carried into the body cavity to a point where the first suture section **542** intersects and traverses the generally planar opening defined by the expanded snare loop **534**.

[0147] The guide **530** is then tilted in the opposite direction, such that the opposite second lateral tip **536** of the second snare loop **534** comes in close proximity to the peritoneum **540**. With the tilted guide **530** in position, the second free end section of suture **541** may then be engaged by the suture engaging device **501**, and inserted through the opposing channel to place the second end suture section **543**. Once the needle **551** exits the opposing channel it passes through the compressed layers of tissue **539**, **540** and enters the body cavity. As it enters the body cavity, the needle **551** passes through the opposing second snare loop **534**. The needle **551** may release the strand of suture **541** and be removed from the body leaving the suture sections **542**, **543** within the boundaries of the respective snare loops **534**.

[0148] By tilting the guide **530** such that the lateral tip **536** of each receiving snare loop **534** comes into contact with the adjacent peritoneum **540**, a greater and more desirable horizontal distance, or "bite," is achieved between each puncture location of the needle **501** through the peritoneum **540** to the closest edge of the defect coinciding with the sidewall of the barrel.

[0149] Moving the actuator section **533** away from the barrel retracts the snare loops **534** so as to capture the suture sections **542**, **543**. With both suture sections **542**, **543** captured, the guide **530** is retracted from the tissue track, carrying the suture sections **542**, **543** and directing the suture in the inner abdominal space to create a stitch. With the guide **530** and suture sections **542**, **543** exposed outside the body cavity, the actuator section **533** can be pushed toward the barrel **531** extending the suture snare loops **534**, and thus releasing the suture sections **542**, **543**. A knot can then be tied in the suture sections **542**, **543** and secured to provide closure of the wound.

[0150] FIG. **30** illustrates a preferred method **600** for closing a surgical wound using suture. Step **610** comprises actuating a guide into a slender configuration for insertion into a body cavity by retracting a monolithic snare. Step **620** comprises translating the monolithic snare distally with respect to a barrel to deploy at least two suture snare loops so as to radially expand a securing mechanism against an internal wall of the body cavity. Step **620** comprises positioning both channel exits beneath the outer surface of the skin.

[0151] If a thinner abdominal wall is involved, then optional step **630** may be employed by tilting the guide in a first direction to cause a distal tip of a first snare loop to approach or abut an inner peritoneal wall.

[0152] Step **640** comprises engaging a first section of suture with a suture engaging device, inserting the suture engaging device into a first channel of a guide, releasing the first section of suture in the body cavity and retracting the device from the first channel.

[0153] Again, if a thinner abdominal wall is involved, then optional step **650** may be employed by tilting the guide in a second direction opposite the first direction to cause a distal tip of a second snare loop to approach or abut the inner peritoneal wall.

[0154] Step **660** comprises engaging a second section of suture with the suture engaging device, inserting the suture engaging device into a second channel of the guide, releasing the second section of suture in the body cavity at an equiangular spacing from the released first section of suture, and retracting the device from the second channel. In a preferred embodiment where only two snare loops are provided, then the channel exits and associated snare loops are positioned 180 degrees from each other.

[0155] Step **670** comprises actuating the monolithic snare to retract the snare loops and secure the first and second sections of suture while forming a slender configuration of the guide. Step **670** may simply involve translating the actuation section in a proximal direction with respect to the barrel.

[0156] Step **680** comprises retracting the guide in the slender configuration with the captured suture from the body.

[0157] This method **600** may also comprise rotating the guide and repeating steps **610** through **680** to form as many stich loops as desired.

[0158] Referring to FIGS. **33**A, **33**B, **34**A, **34**B and **34**C, a preferred method for closing a surgical wound includes using a guide **700** with markers **702** along the guide body **704**. The markers **702** could be a graphic of graduated lines, a graphic logo, a distinct geometric feature, or a variety of other indicators. FIG. **33**B illustrates graduated lines as markers **702** on a guide.

[0159] The guide **700** has channels **706** for a suture passing needle **708** as described above. The angulation of the channels **706** directs the suture passing needle **708** through a determined path. The intersection of the path of the suture passing needle **708** through tissue **710** is determined by the vertical position of the guide **700** within the tissue **710**. Aligning the markers **702** on the guide **700** with the inner abdominal wall **712** (i.e. peritoneum), provides an active means to predictably pass the suture passing needle **708** through the tissue **710** at a determined distance (D**1**, D**2**, D**3**) from the edge **714** of the fascial defect, or in other terms, the fascial bite of tissue. FIG. **34**A illustrates the "8" graphic as a marker on the guide **700** being aligned with the peritoneum **712** to achieve an 8 mm bite of fascia. FIG. **34**B illustrates the "5" graphic as a marker on the guide **700** being aligned with the peritoneum **712** to achieve a 5 mm bite of fascia. FIG. **34**C illustrates the snare exit **720** ("11") as a marker on the guide being aligned with the peritoneum **712** to achieve a 11 mm bite of fascia.

[0160] FIGS. **35**A and **35**B illustrate how a conventional suture guide, that is positioned relative to the skin surface, may result in too large of a bite of fascia when the patient has a distance from skin **800** to fascia **802** of 60 mm (FIG. **35**A) or too little of a bite of fascia when the patient has a distance from the skin **800** to the fascia **802** of 35 mm (FIG. **36**B). To the contrary, as shown in FIGS. **36**A and **36**B, the suture guide system of the present invention, that is positioned relative to the peritoneum, can be used, as discussed above, to align the guide to the proper depth each time, ensuring a constant bite of fascia (such as 8 mm, as illustrated in the Figures, however, other desired bites can be used, depending on the specifics of the suture, the patent, or the like).

[0161] The fascia bite created when closing a tissue defect can have significant clinical impact. A large bite of fascia will result in excess tissue within the suture loop stitch and when the stitch is tied and the knot tensioned, the tissue is prone to be strangulated and could cause significant post operative pain for the patient at the defect sight, especially for younger patients with healthy tissue. Not enough bite of fascia can result in the suture tearing through the small bite of fascia resulting in an "air knot", requiring replacement of the stitch, which is more susceptible for older patients with flaccid tissue. The ability to adjust the fascial bite to meet the clinical need of the particular patient is an advantageous clinical benefit.

[0162] In some embodiments, the markings **702** may extend along a portion of the guide, as shown. In other embodiments, the markings **702** may extend from the snare opening **720** upward along the length of the guide **700**.

[0163] All the features disclosed in this specification, including any accompanying abstract and drawings, may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

[0164] Claim elements and steps herein may have been numbered and/or lettered solely as an aid in readability and understanding. Any such numbering and lettering in itself is not intended to and should not be taken to indicate the ordering of elements and/or steps in the claims.

[0165] Many alterations and modifications may be made by those having ordinary skill in the art without departing from the spirit and scope of the invention. Therefore, it must be understood that the illustrated embodiments have been set forth only for the purposes of examples and that they should not be taken as limiting the invention as defined by the following claims. For example, notwithstanding the fact that the elements of a claim are set forth below in a certain combination, it must be expressly understood that the invention includes other combinations of fewer, more or different ones of the disclosed elements.

[0166] The words used in this specification to describe the invention and its various embodiments are to be understood not only in the sense of their commonly defined meanings, but to include by special definition in this specification the generic structure, material or acts of which they represent a single species.

[0167] The definitions of the words or elements of the following claims are, therefore, defined in this specification to not only include the combination of elements which are literally set forth. In this sense it is therefore contemplated that an equivalent substitution of two or more elements may be made for any one of the elements in the claims below or that a single element may be substituted for two or more elements in a claim. Although elements may be described above as acting in certain combinations and even initially claimed as such, it is to be expressly understood that one or more elements from a claimed combination can in some cases be excised from the combination and that the claimed combination may be directed to a subcombination or variation of a subcombination.

[0168] Insubstantial changes from the claimed subject matter as viewed by a person with ordinary skill in the art, now known or later devised, are expressly contemplated as being equivalently within the scope of the claims. Therefore, obvious substitutions now or later known to one with ordinary skill in the art are defined to be within the scope of the defined elements.

[0169] The claims are thus to be understood to include what is specifically illustrated and described above, what is conceptually equivalent, what can be obviously substituted and also what incorporates the essential idea of the invention.

1-16. (canceled)

17. A method of controlling a fascial bite of tissue when closing a trocar site, comprising:

inserting a surgical guide device into the trocar site, the surgical guide device comprising a barrel defining a first channel and a second channel, the first channel comprising a first exit for directing a needle into a body cavity, the second channel comprising a second exit for directing the needle into the body cavity;

visualizing a plurality of numerical markings disposed along an exterior surface of the barrel;

aligning a selected one of the plurality of numerical markings with an inner abdominal wall;

inserting the needle through the first channel of the barrel to guide a suture;

inserting the needle through the second channel, opposite the first channel of the barrel, to guide the suture; and

utilizing a distal tip of the barrel, when positioned into the body cavity, to direct the suture to create a stitch, wherein

the fascial bite of tissue is set based on the selected one of the plurality of numerical markings that are aligned with the inner abdominal wall.

18. The method of claim **17**, wherein the fascial bite of tissue is selected based on a clinical need of a patient.

19. The method of claim **17**, wherein the selected one of the plurality of numerical markings is selected to prevent excess tissue within a suture loop stitch.

20. The method of claim **17**, wherein the selected one of the plurality of numerical markings is aligned with an inner abdominal wall to predictably pass the needle through the first channel or the second channel and through tissue at a determined distance from an edge of a fascial defect.

21. The method of claim **17**, wherein the plurality of numerical markings include a logo.

22. The method of claim **17**, wherein the plurality of numerical markings include a plurality of lines.

23. The method of claim **17**, further comprising:

moving first and second self-supporting suture catchers in unison to the retracted configuration by moving an actuator proximally with respect to the barrel; and

moving the first and second self-supporting suture catchers in unison to the radially extended configuration by moving the actuator distally with respect to the barrel.

24. The method of claim **23**, further comprising extending the first self-supporting suture catcher and the second self-supporting suture catcher radially outward and in opposite directions from the distal tip without the distal tip axially shortening with respect to the barrel when changing between the retracted configuration and the radially extended configuration.

25. The method of claim **23**, further comprising retracting the first self-supporting suture catcher and the second self-supporting suture catcher in unison to close a first opening and a second opening while simultaneously capturing a first suture portion and a second suture portion.

26. The method of claim **24**, further comprising:

passing and dropping the first suture portion into the first opening formed by the first self-supporting suture catcher, such that the first suture portion resides loosely within the first opening; and

passing and dropping the second suture portion into the second opening formed by the second self-supporting suture catcher, such that the second suture portion resides loosely.

27. A method of controlling a fascial bite of tissue when closing a trocar site, comprising:

inserting a surgical guide device into the trocar site, the surgical guide device comprising a barrel defining a first channel and a second channel, the first channel comprising a first exit for directing a needle into a body cavity, the second channel comprising a second exit for directing the needle into the body cavity, the barrel having a plurality of numerical markings disposed along an exterior surface thereof,

aligning a selected one of the plurality of numerical markings with an inner surface of tissue through which the surgical guide is inserted;

inserting the needle through the first channel of the barrel to guide a suture;

inserting the needle through the second channel, opposite the first channel of the barrel, to guide the suture; and

directing the suture with a distal member of the surgical guide that is within the abdominal cavity to create a stitch, wherein

the fascial bite of the tissue is set based on the selected one of the plurality of numerical markings that are aligned with the inner surface of tissue.

28. The method of claim **27**, wherein the selected one of the plurality of numerical markings are aligned with the inner surface of tissue through visualization of the selected one of the plurality of the numerical markings within a body cavity.

29. The method of claim **27**, wherein the inner surface of tissue is a peritoneum of an abdominal cavity.

30. The method of claim **27**, wherein the plurality of numerical markings include a logo.

31. The method of claim **27**, wherein the plurality of numerical markings include a plurality of lines.

32. The method of claim **27**, wherein the fascial bite of tissue is selected based on a clinical need of a patient.

33. The method of claim **27**, wherein the selected one of the plurality of numerical markings is selected to prevent excess tissue within a suture loop stitch.

34. The method of claim **27**, wherein the selected one of the plurality of numerical markings is aligned with the inner tissue surface to predictably pass the needle through the first channel or the second channel and through tissue at a determined distance from an edge of a fascial defect.

35. A method of controlling a fascial bite of tissue when closing a trocar site, comprising:

inserting a surgical guide device into the trocar site, the surgical guide device comprising a barrel defining a first channel and a second channel, the first channel comprising a first exit for directing a needle into a body cavity, the second channel comprising a second exit for directing the needle into the body cavity, the barrel having a plurality of numerical markings disposed along an exterior surface thereof,

aligning a selected one of the plurality of numerical markings with an inner abdominal cavity surface through which the surgical guide is inserted;

inserting the needle through the first channel of the barrel to guide a suture;

inserting the needle through the second channel, opposite the first channel of the barrel, to guide the suture; and

directing the suture with a distal member of the surgical guide that is within the abdominal cavity to create a stitch, wherein

the fascial bite of the tissue is set based on the selected one of the plurality of numerical markings that are aligned with the inner abdominal cavity surface;

the fascial bite of tissue is selected based on a clinical need of a patient;

the selected one of the plurality of numerical markings is selected to prevent excess tissue within a suture loop stitch; and

the selected one of the plurality of numerical markings is aligned with the inner tissue surface to predictably pass the needle through the first channel or the second channel and through tissue at a determined distance from an edge of a fascial defect.

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